

BFS Traversal

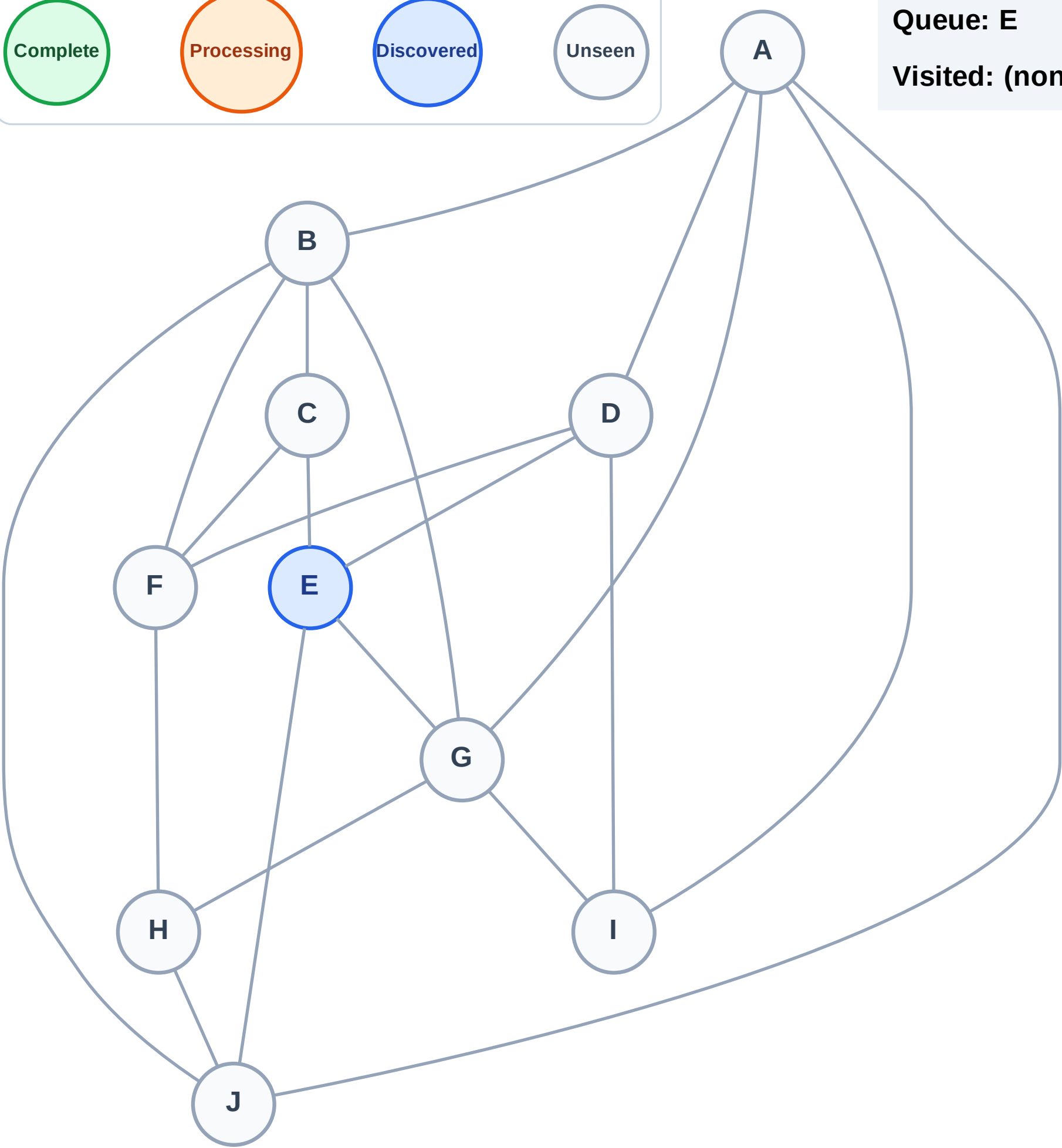
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

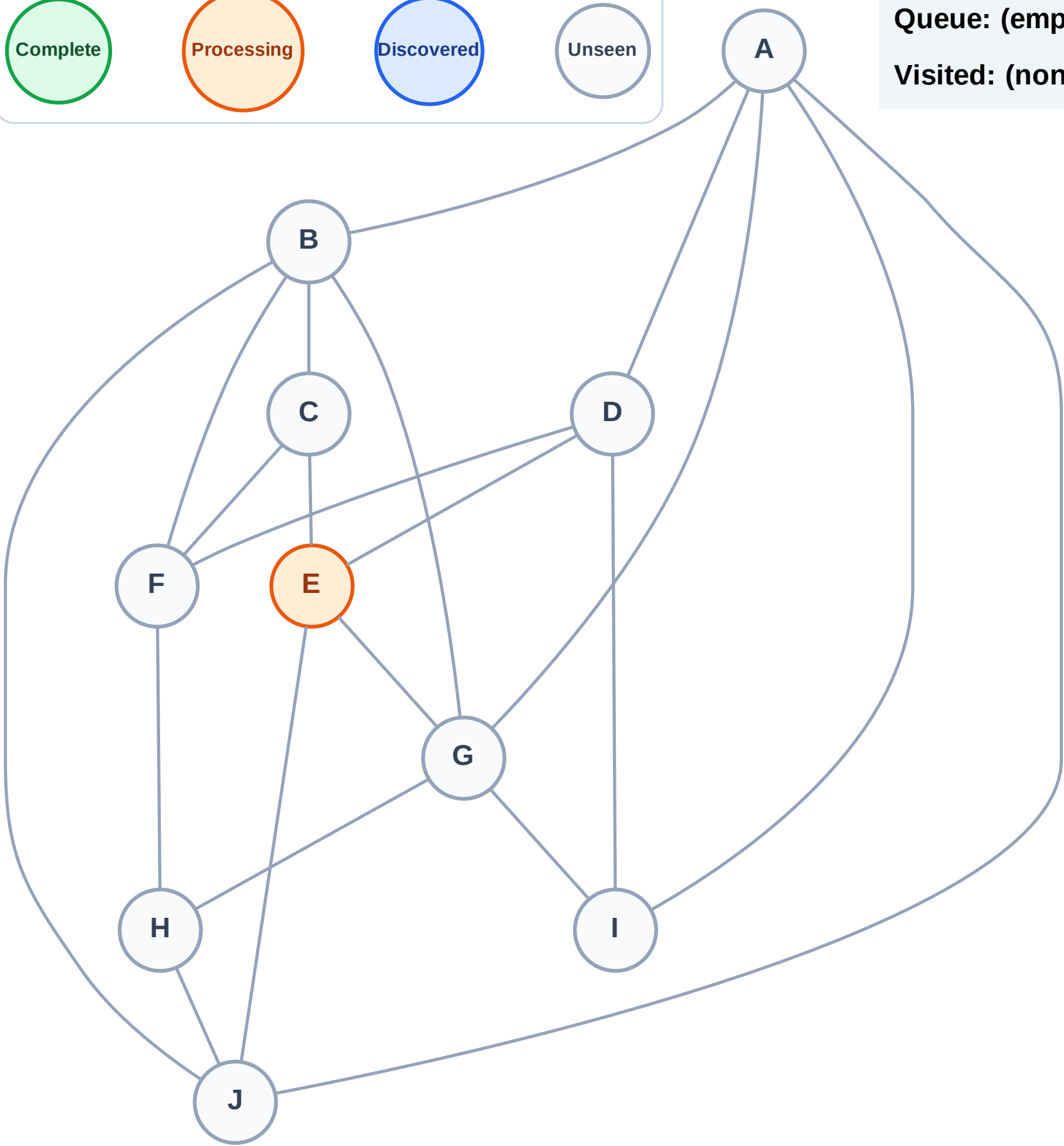
Step 2: Process node E

Legend



Queue: (empty)

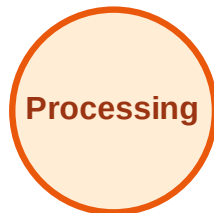
Visited: (none)



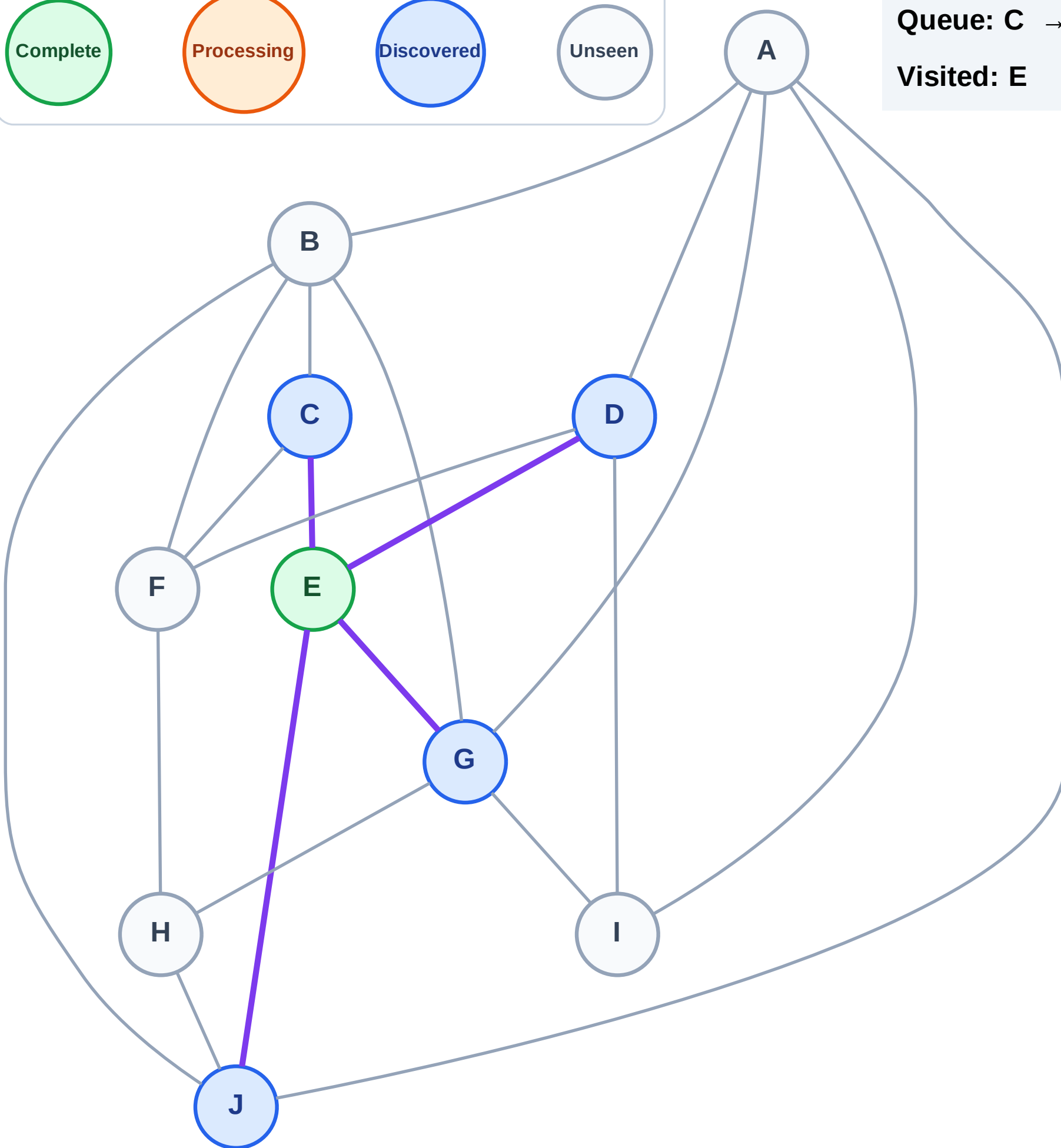
BFS Traversal

Step 3: Discover C, D, G, J from E

Legend



Queue: C → D → G → J
Visited: E



BFS Traversal

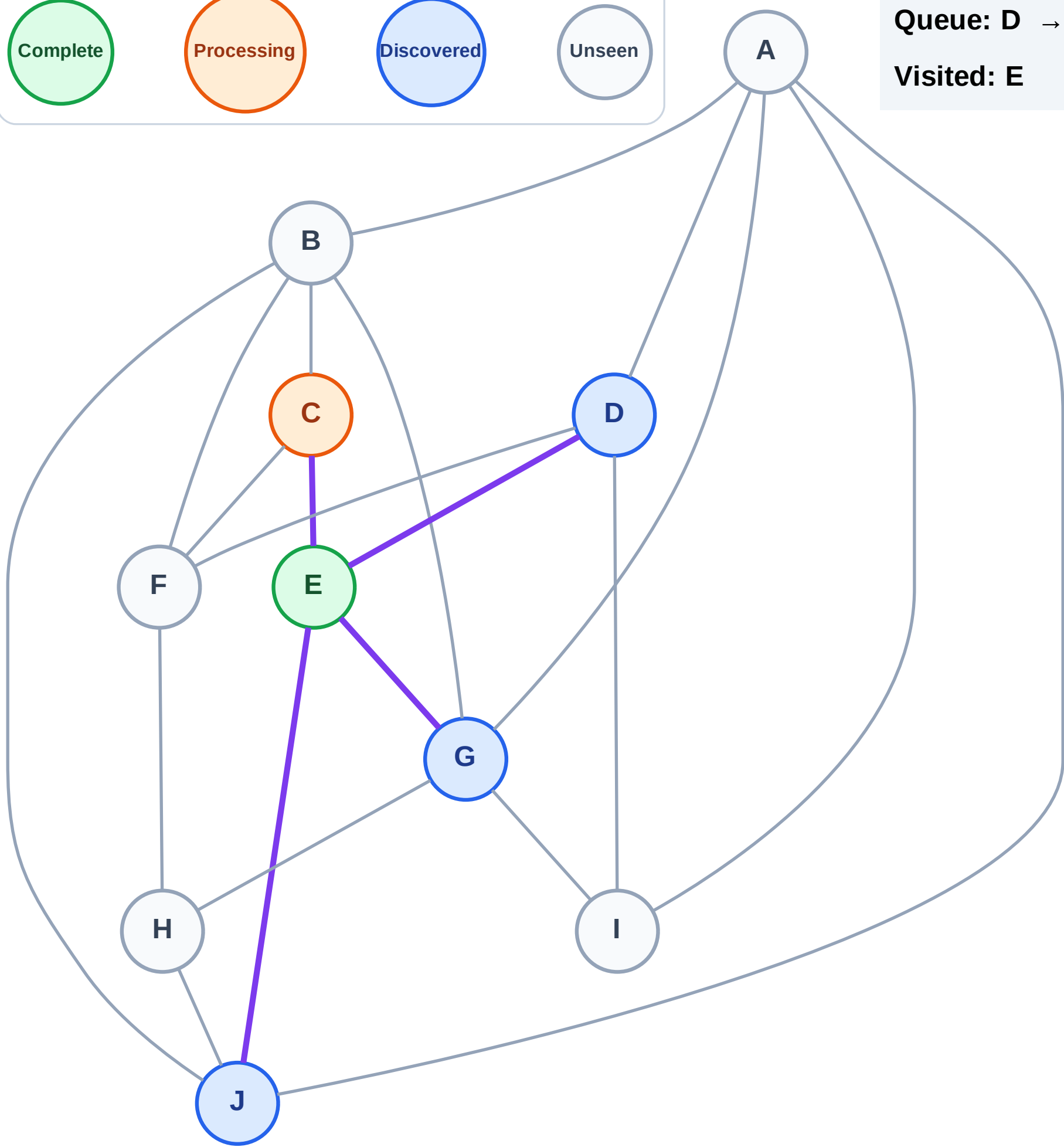
Step 4: Process node C

Legend



Queue: D → G → J

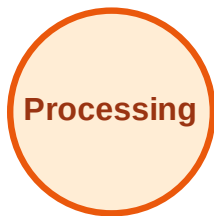
Visited: E



BFS Traversal

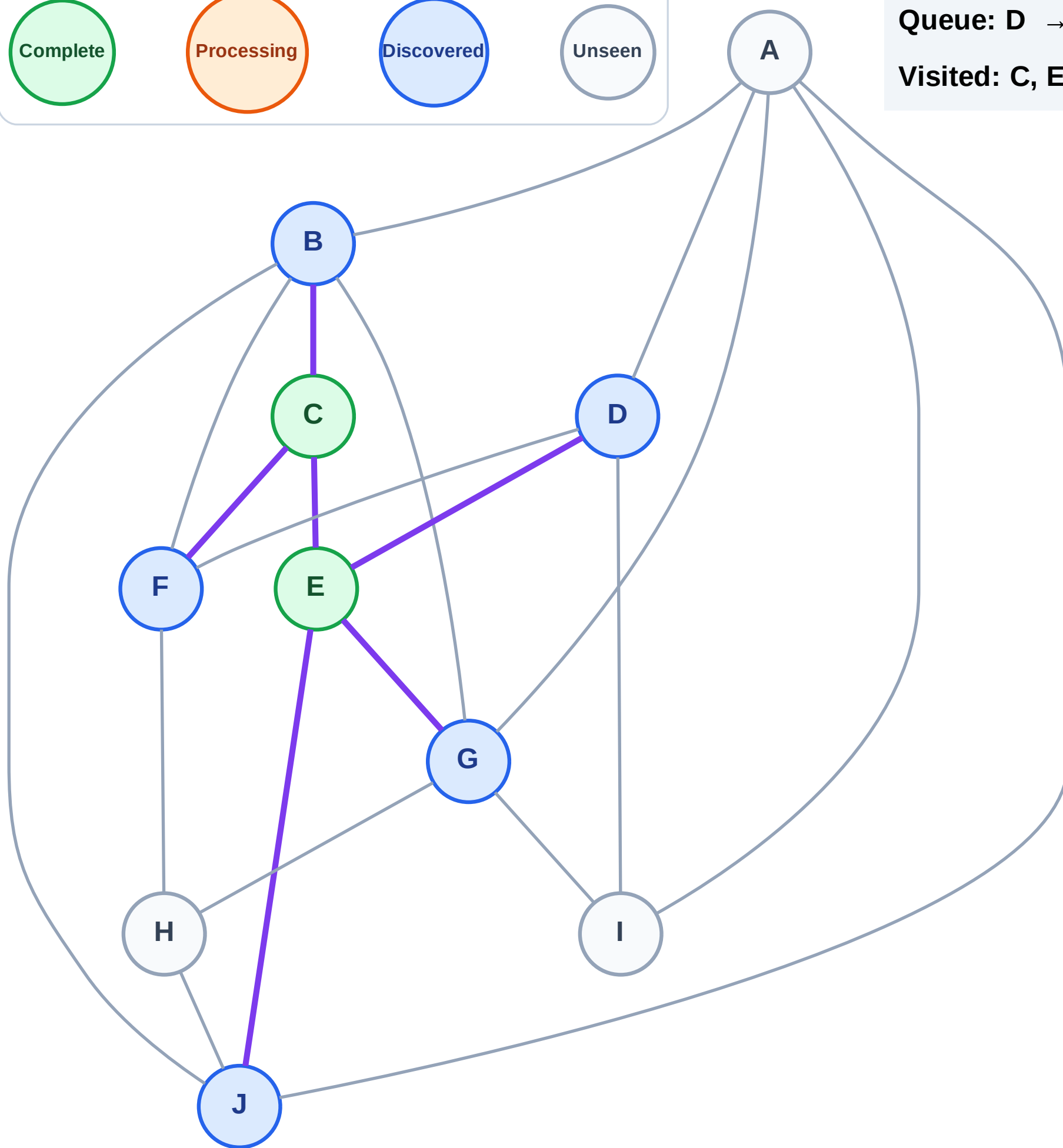
Step 5: Discover B, F from C

Legend



Queue: D → G → J → B → F

Visited: C, E



BFS Traversal

Step 6: Process node D

Legend

Complete

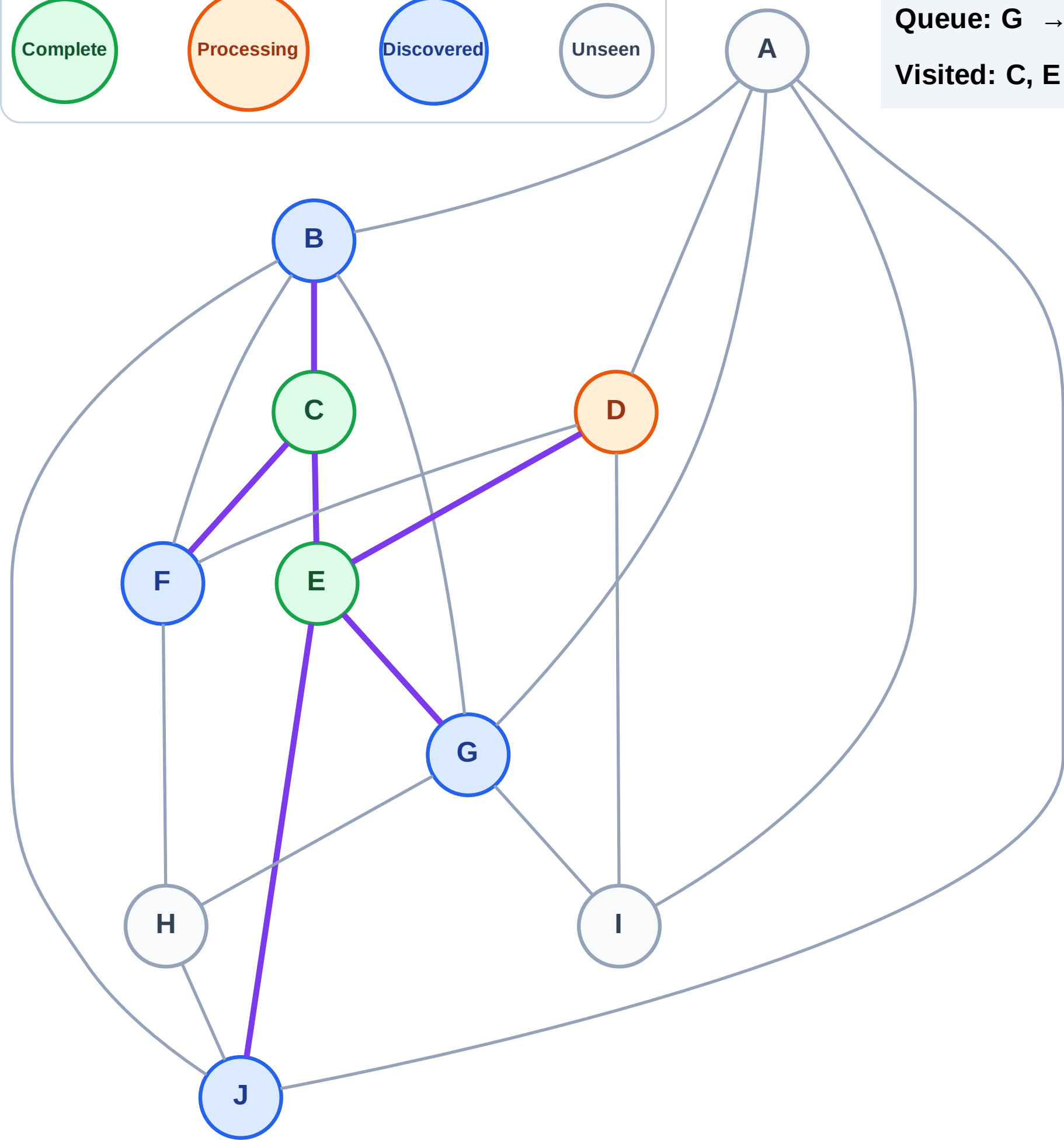
Processing

Discovered

Unseen

Queue: G → J → B → F

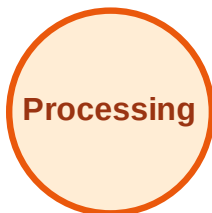
Visited: C, E



BFS Traversal

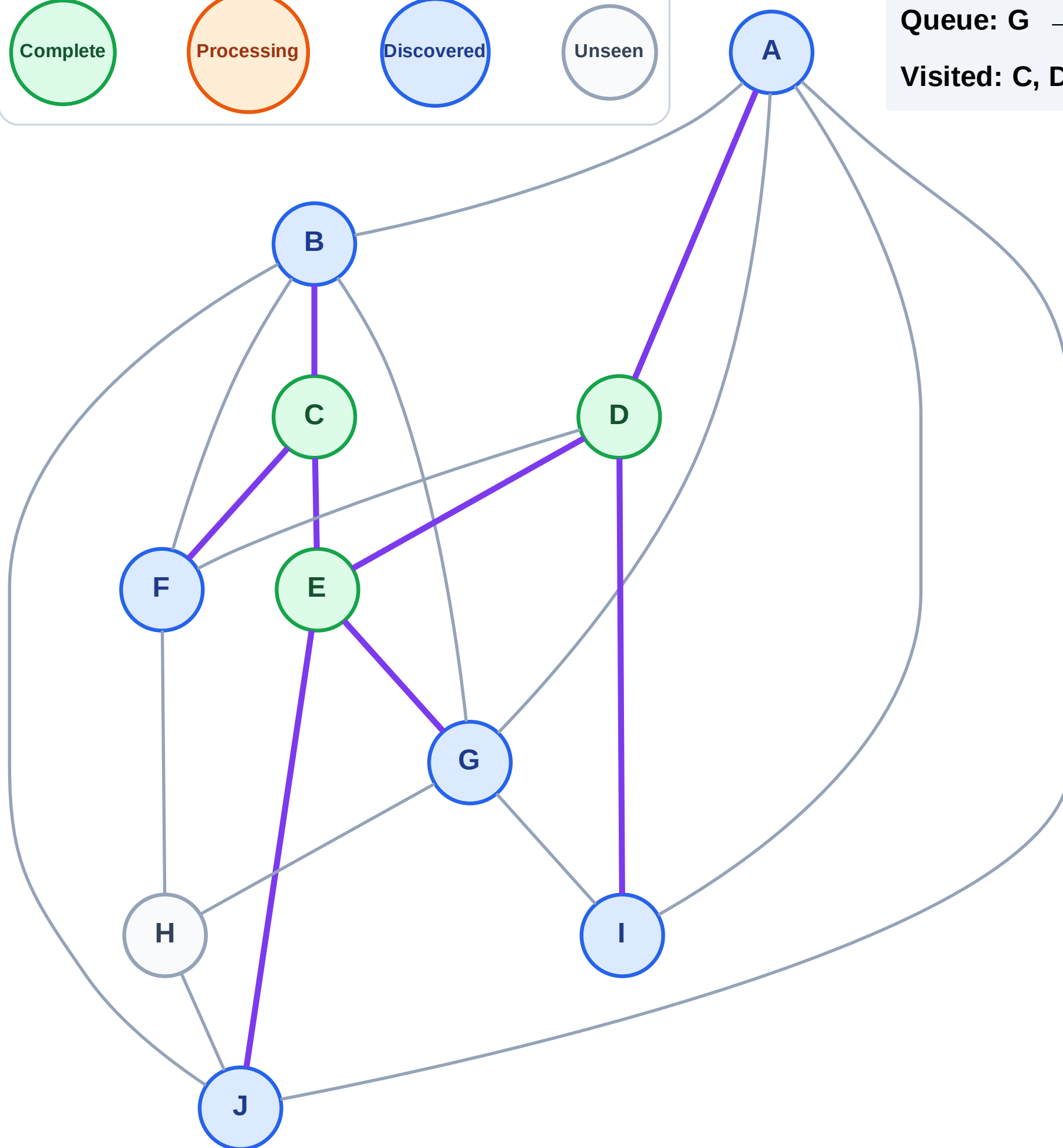
Step 7: Discover A, I from D

Legend



Queue: G → J → B → F → A → I

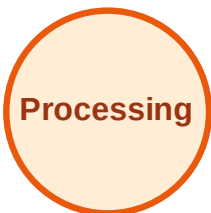
Visited: C, D, E



BFS Traversal

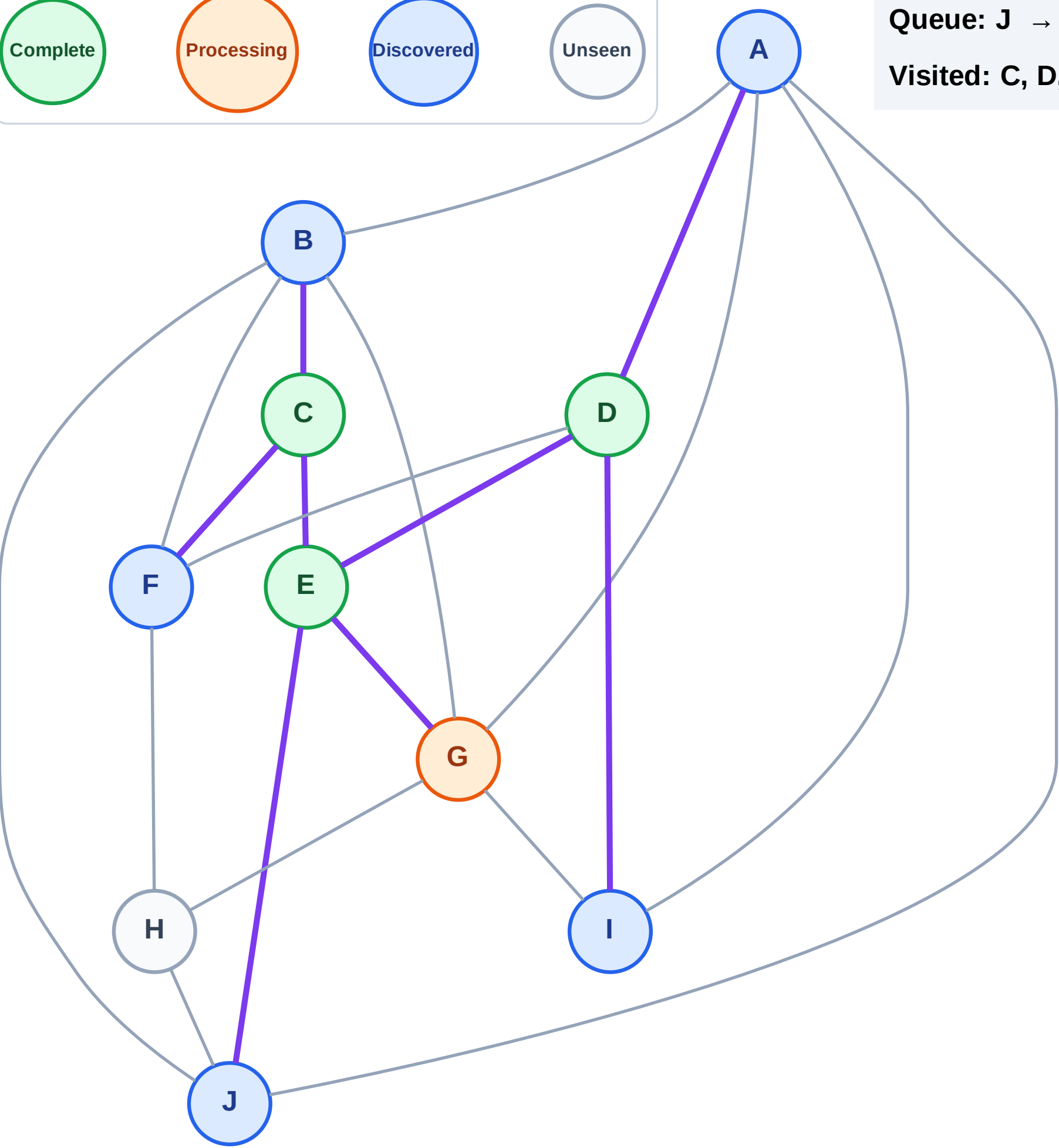
Step 8: Process node G

Legend



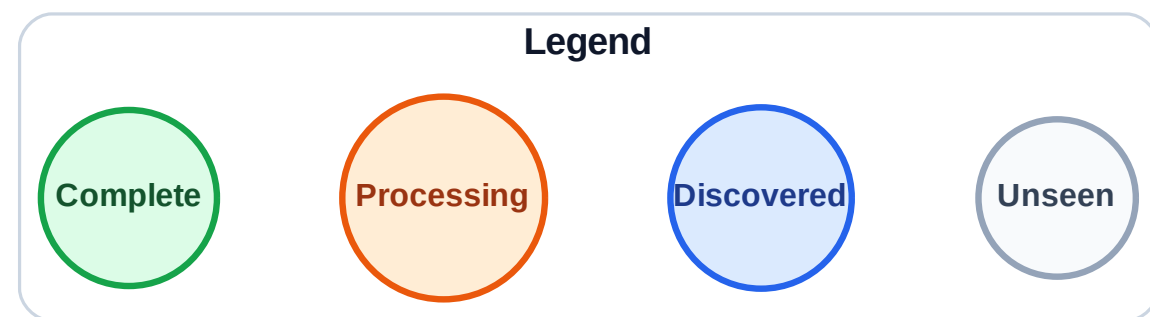
Queue: J → B → F → A → I

Visited: C, D, E



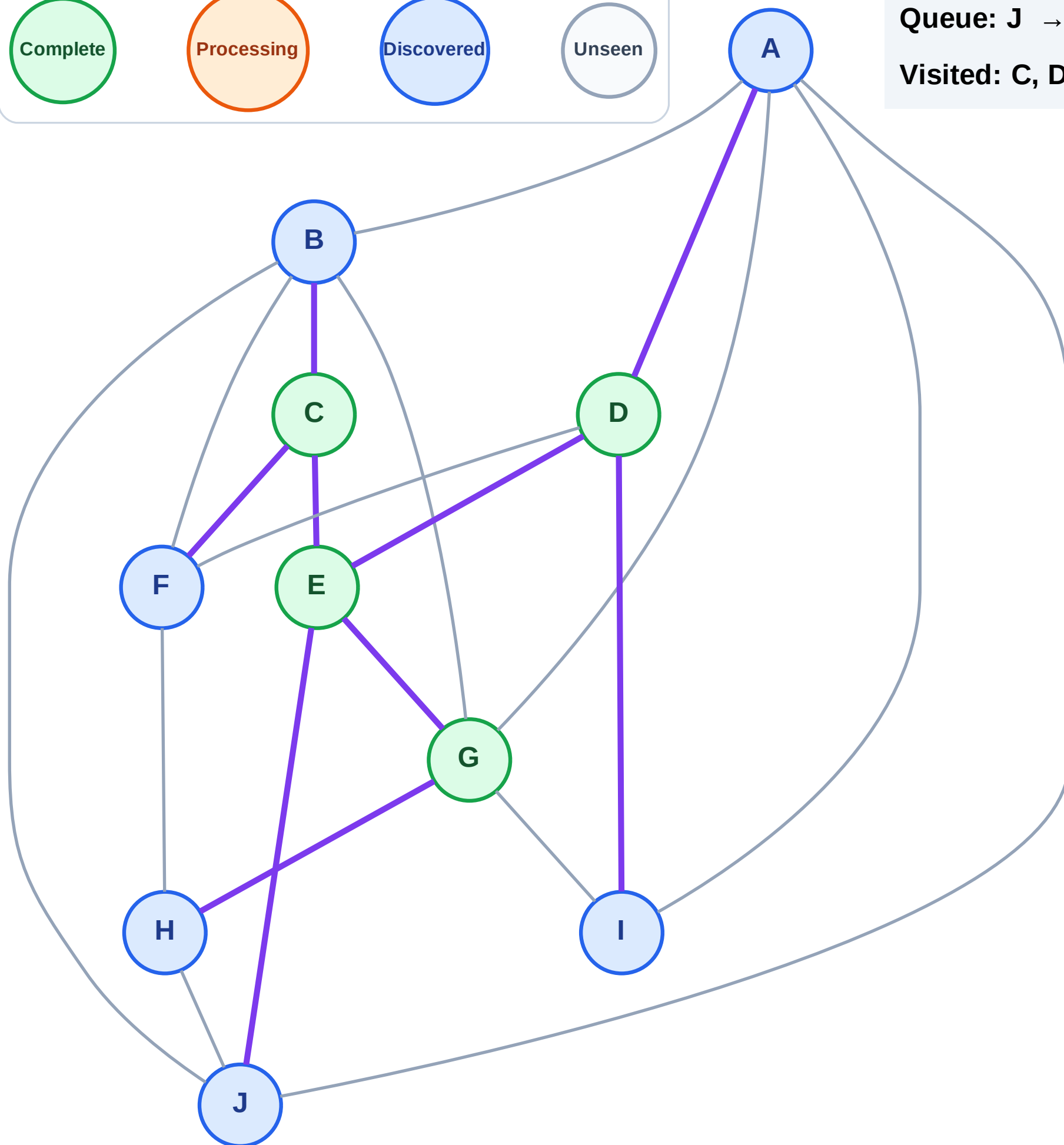
BFS Traversal

Step 9: Discover H from G



Queue: J → B → F → A → I → H

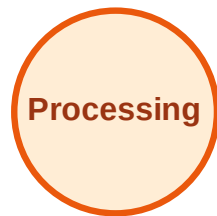
Visited: C, D, E, G



BFS Traversal

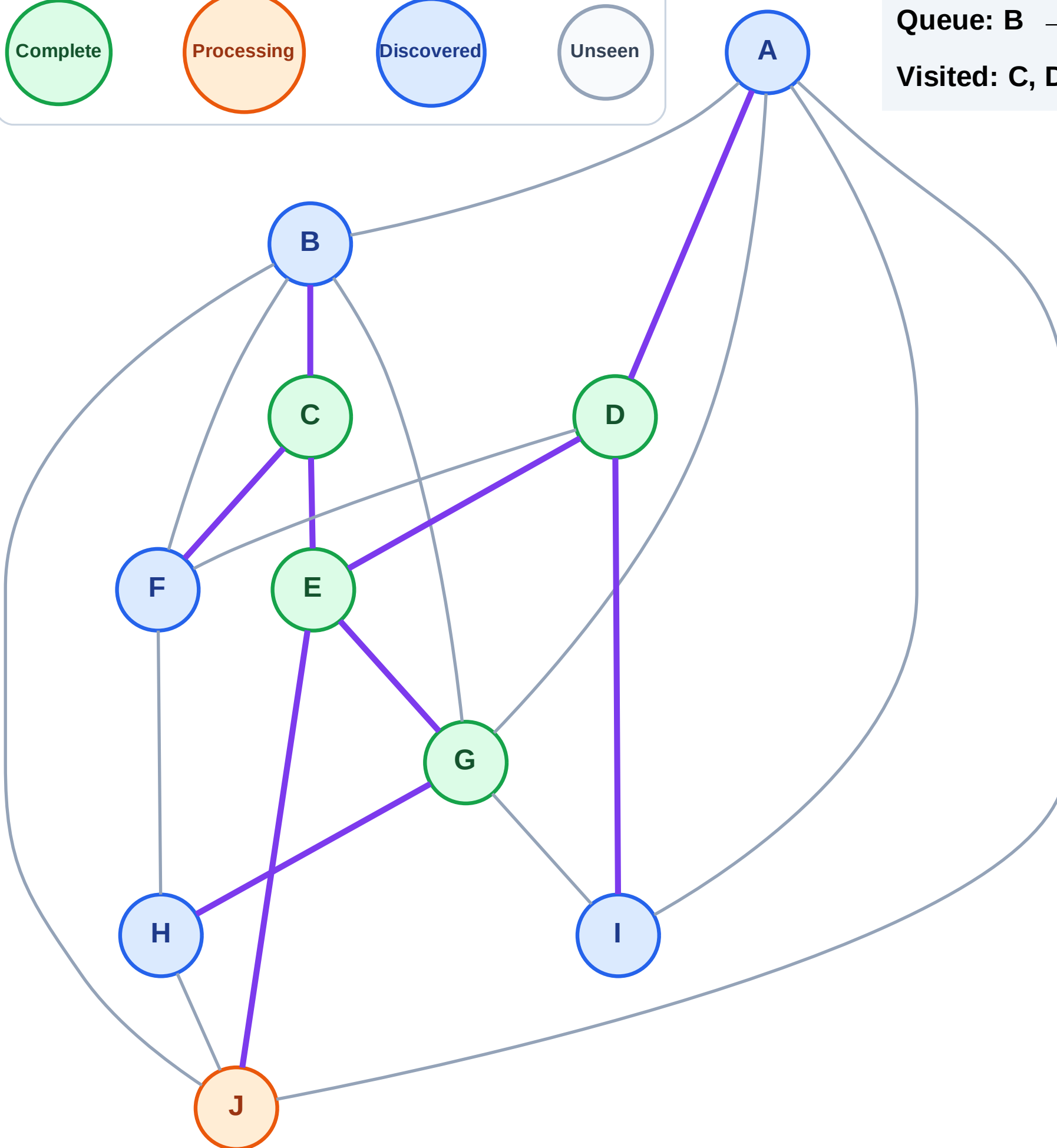
Step 10: Process node J

Legend



Queue: B → F → A → I → H

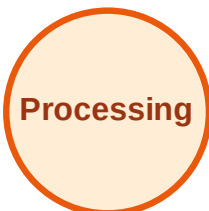
Visited: C, D, E, G



BFS Traversal

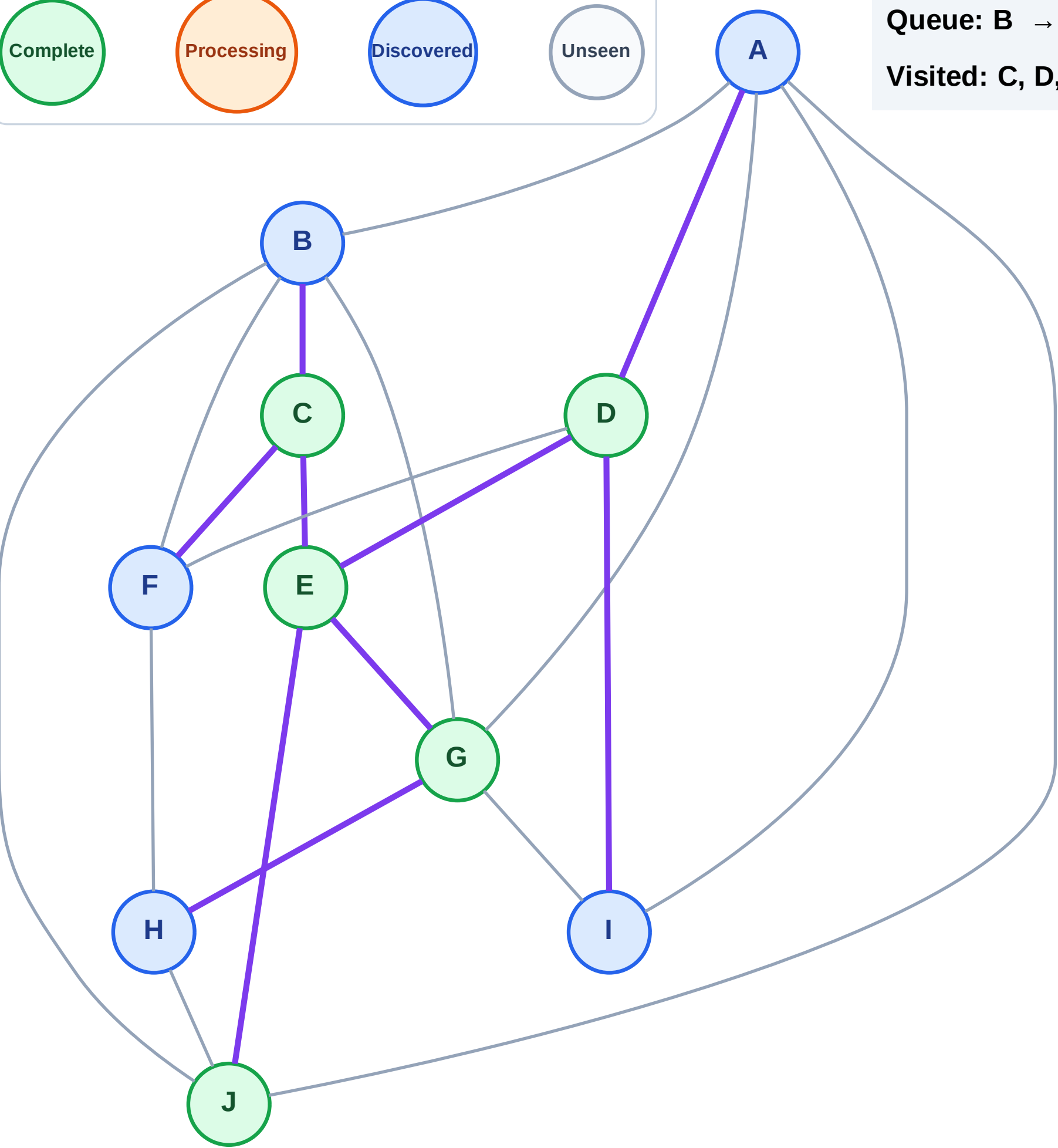
Step 11: No new neighbors from J

Legend



Queue: B → F → A → I → H

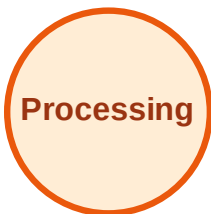
Visited: C, D, E, G, J



BFS Traversal

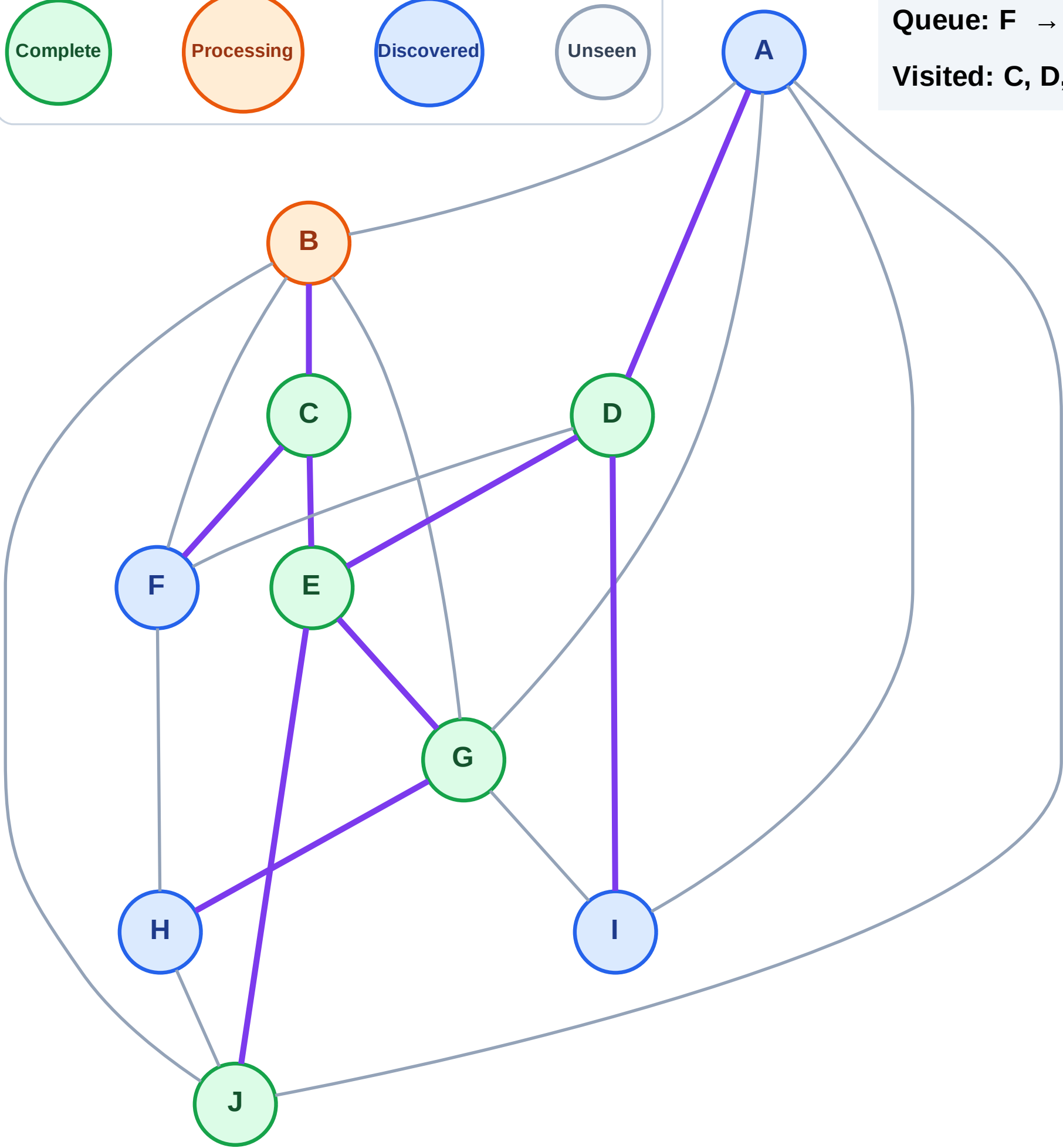
Step 12: Process node B

Legend



Queue: F → A → I → H

Visited: C, D, E, G, J



BFS Traversal

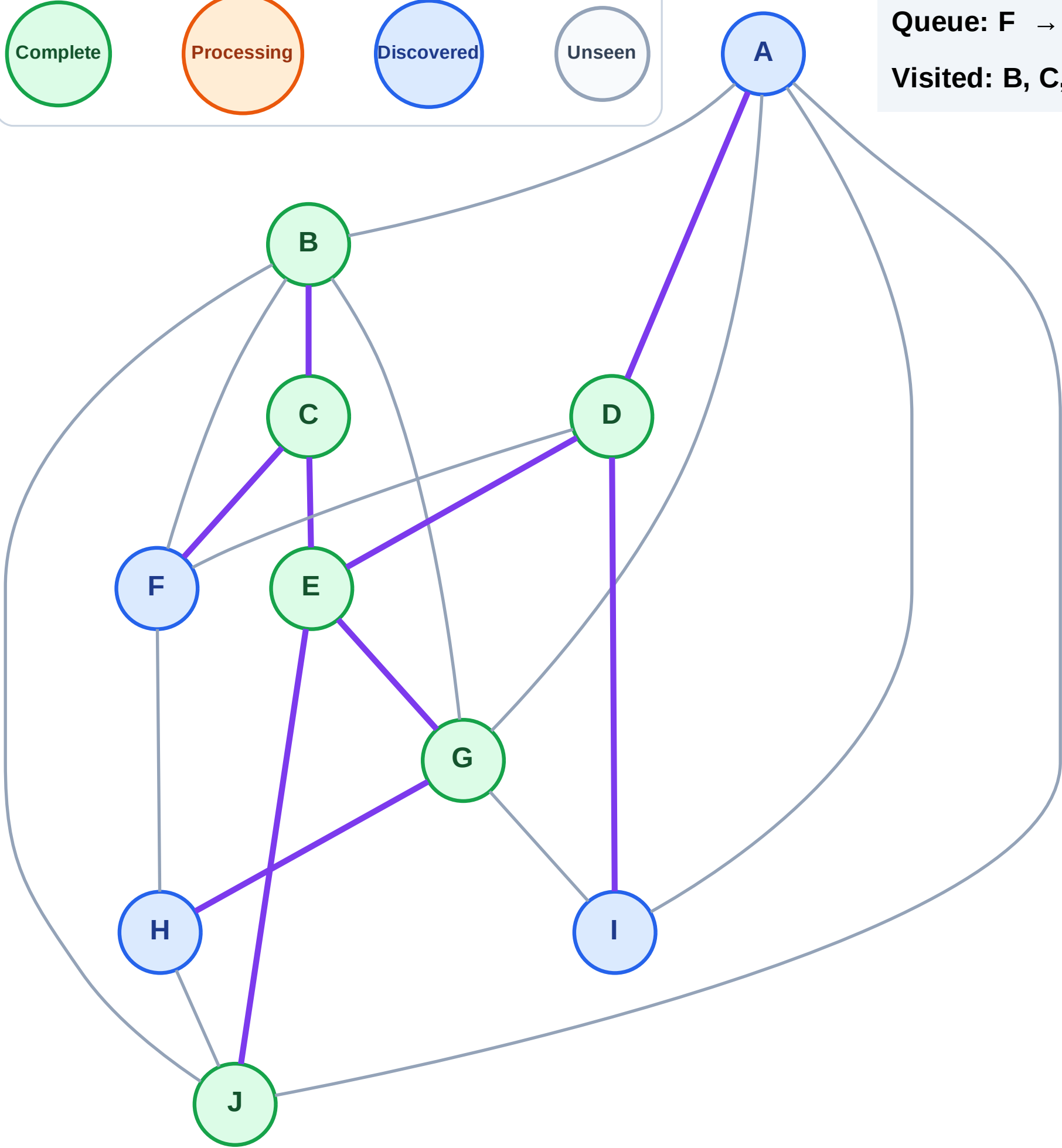
Step 13: No new neighbors from B

Legend



Queue: F → A → I → H

Visited: B, C, D, E, G, J



BFS Traversal

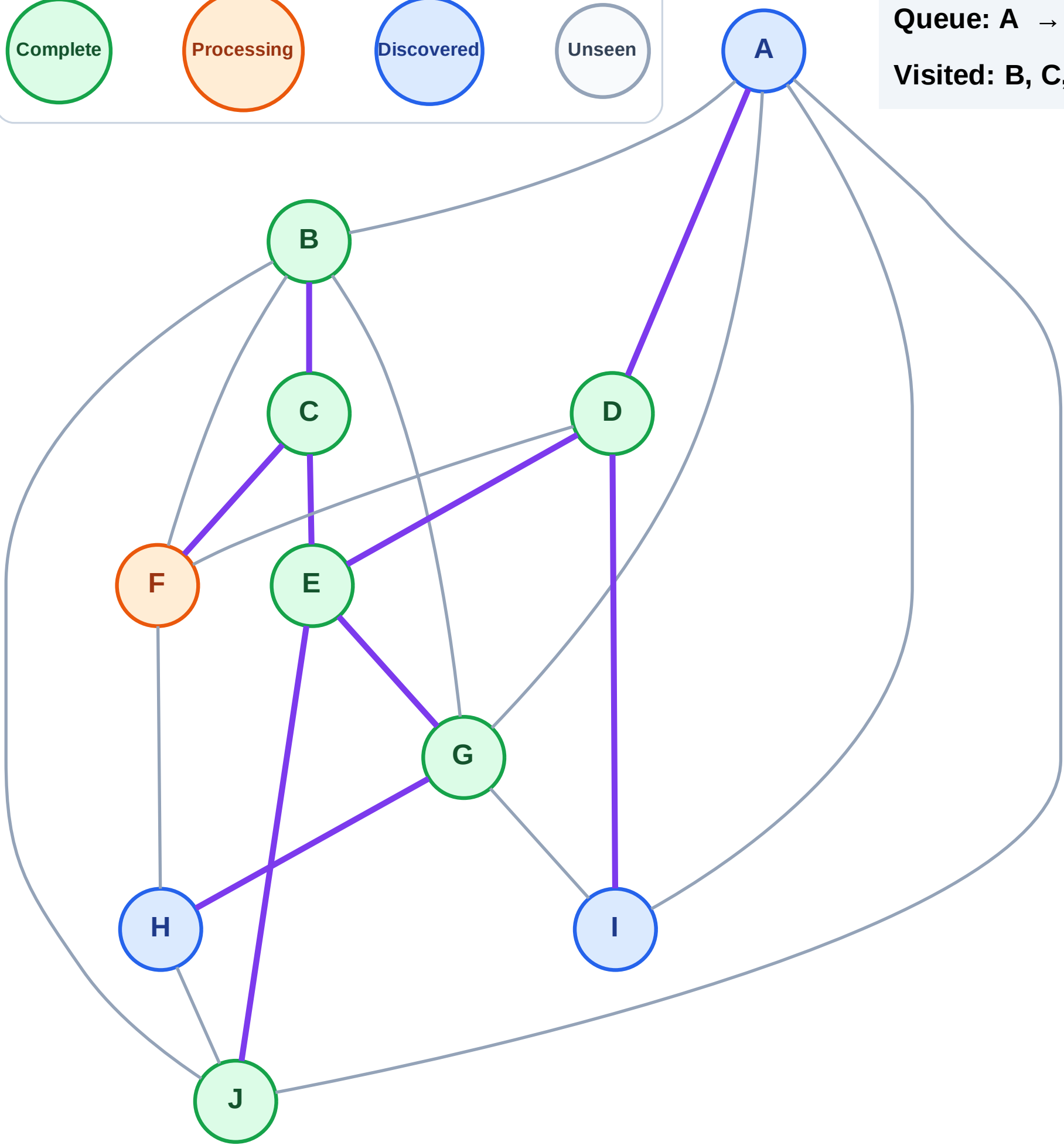
Step 14: Process node F

Legend



Queue: A → I → H

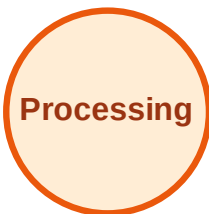
Visited: B, C, D, E, G, J



BFS Traversal

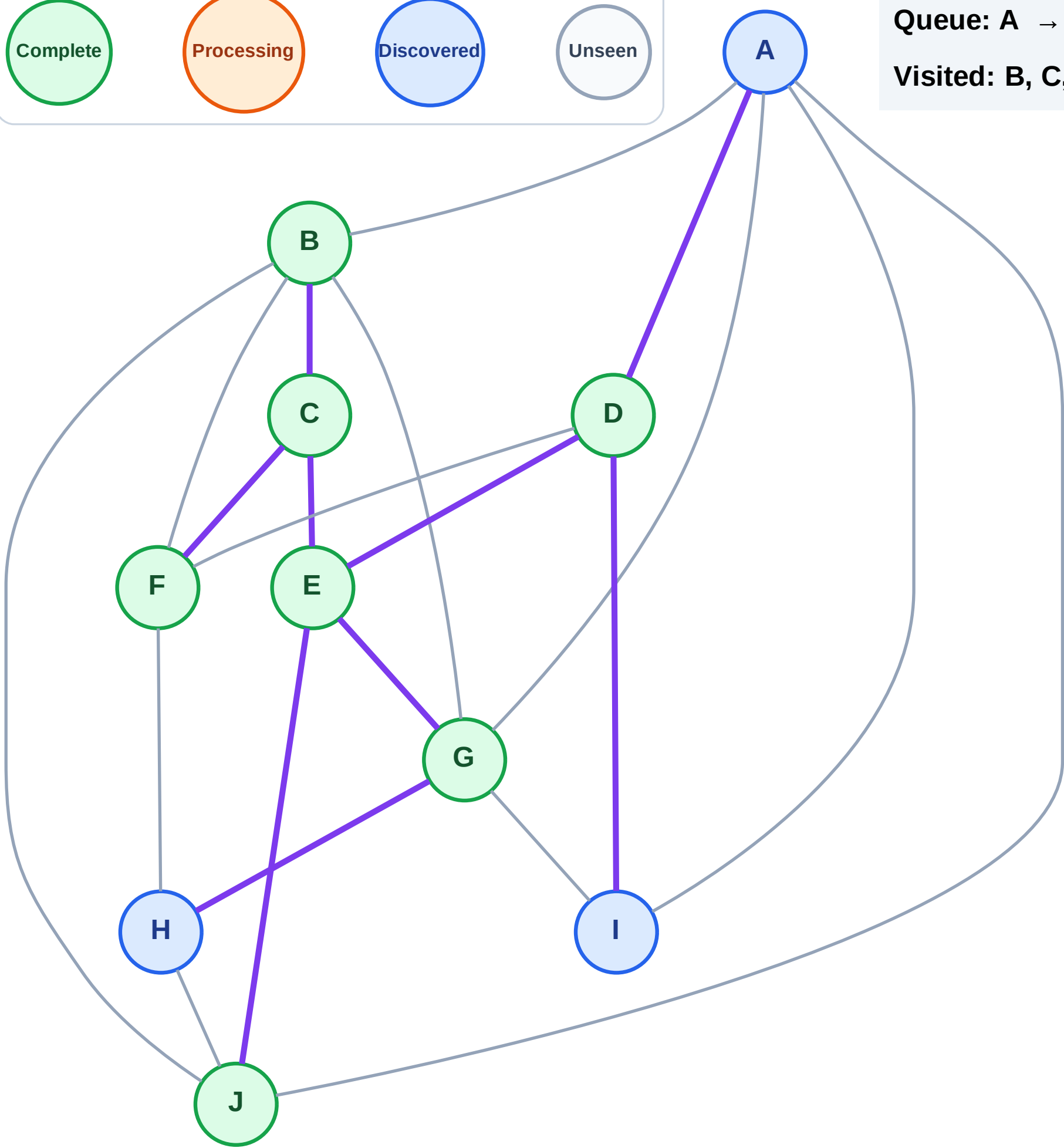
Step 15: No new neighbors from F

Legend



Queue: A → I → H

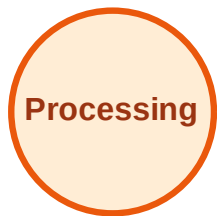
Visited: B, C, D, E, F, G, J



BFS Traversal

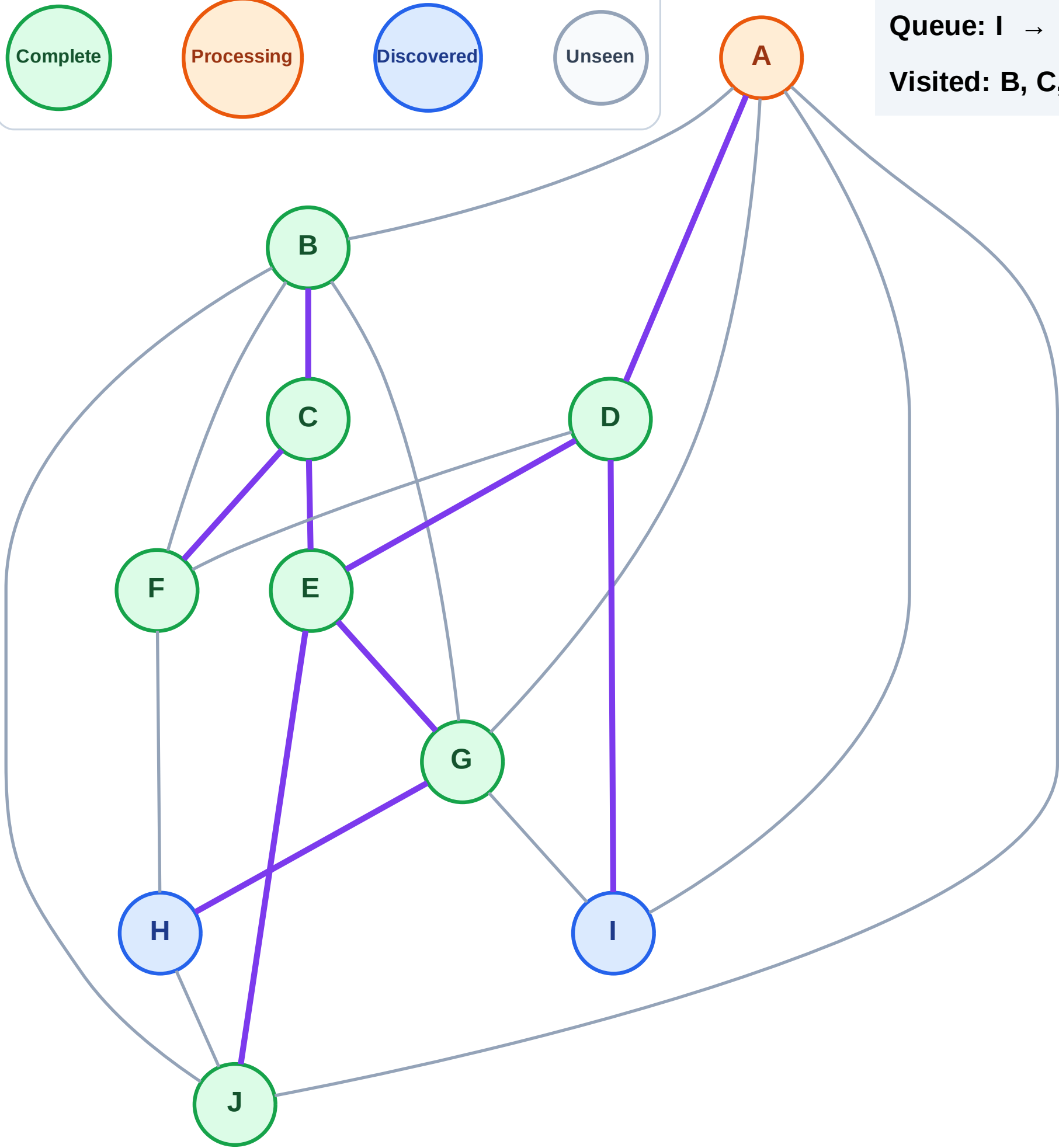
Step 16: Process node A

Legend



Queue: I → H

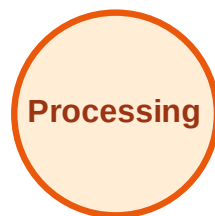
Visited: B, C, D, E, F, G, J



BFS Traversal

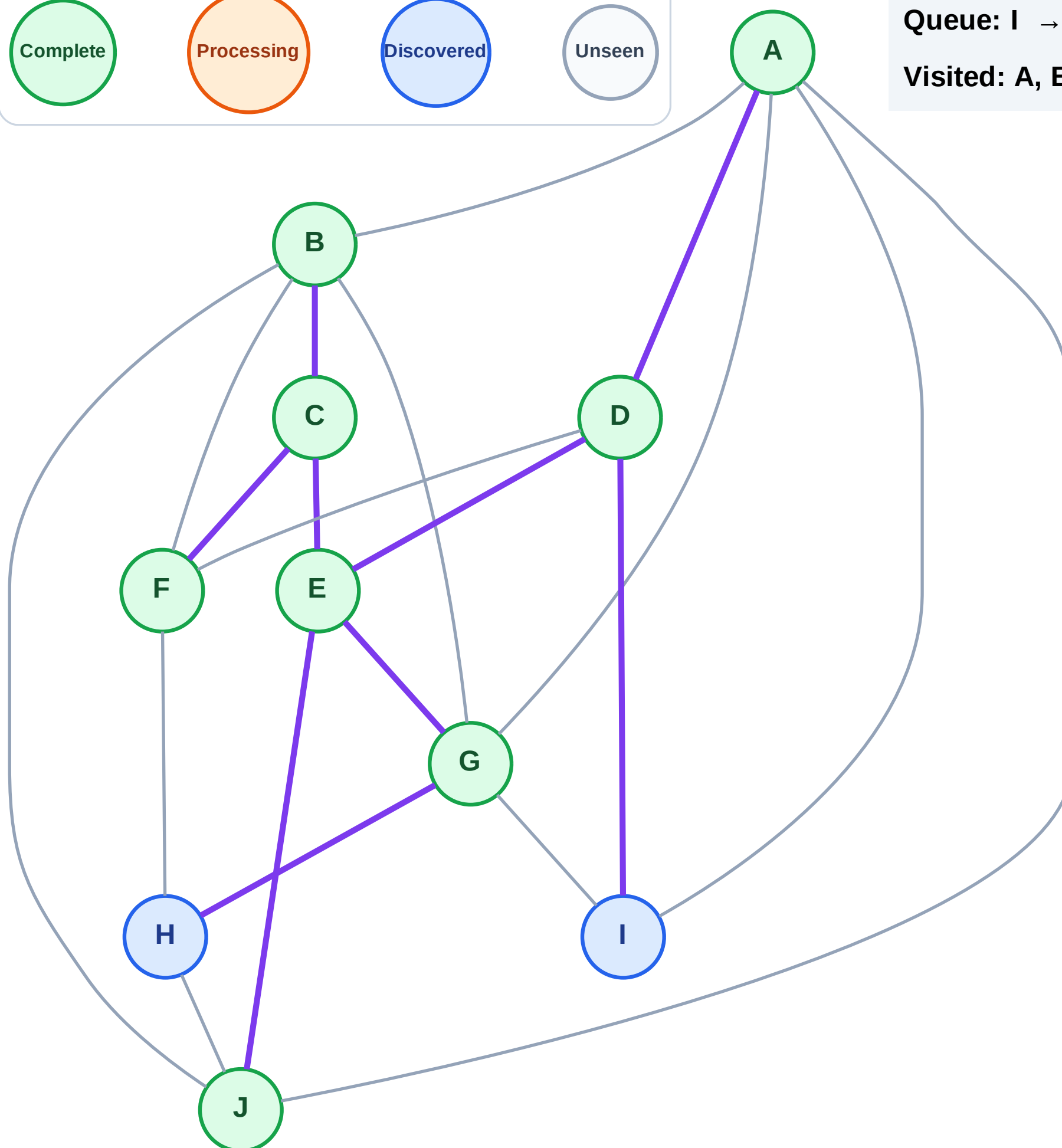
Step 17: No new neighbors from A

Legend



Queue: I → H

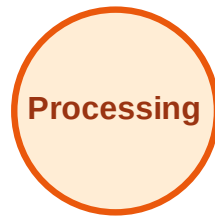
Visited: A, B, C, D, E, F, G, J



BFS Traversal

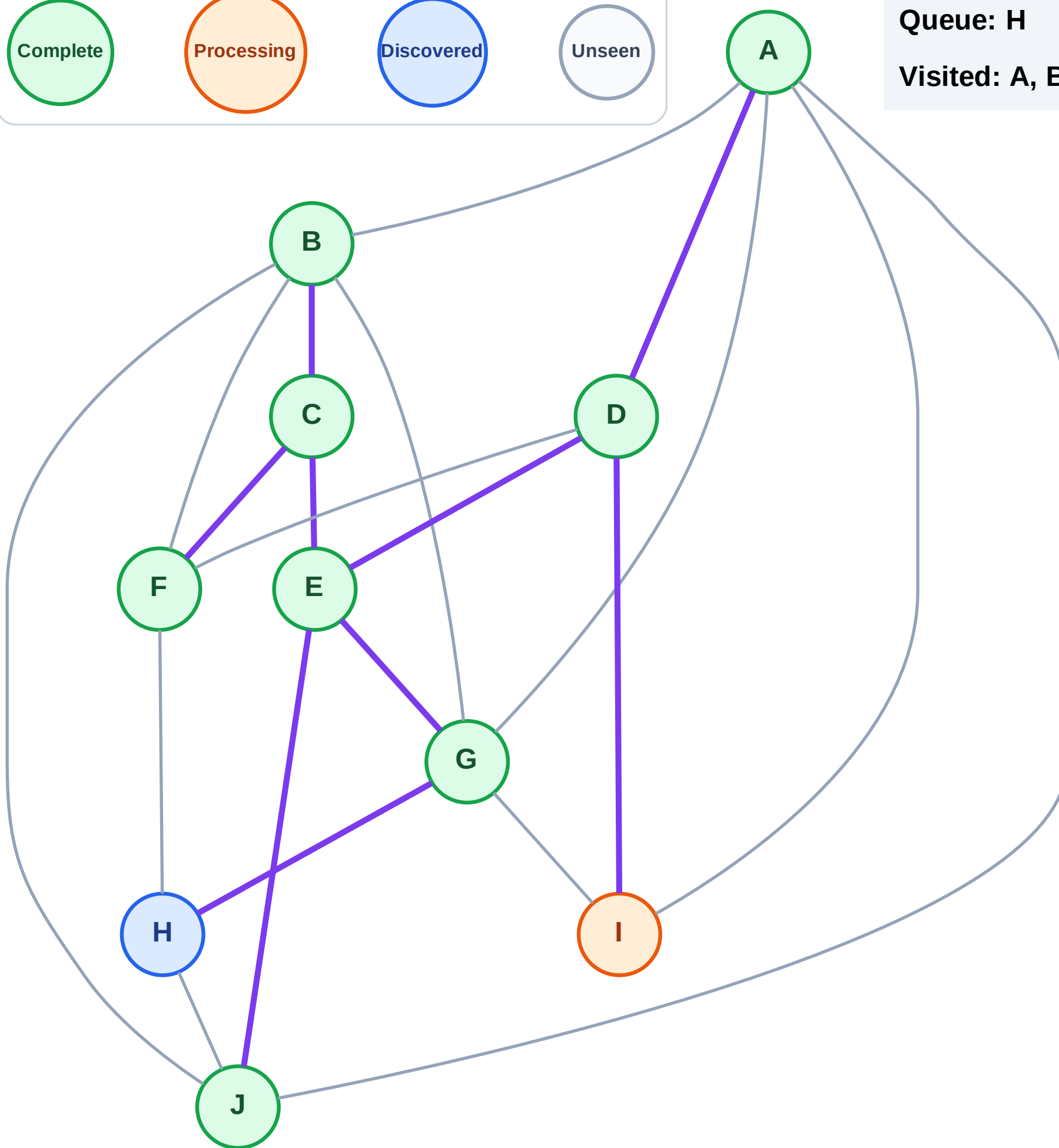
Step 18: Process node I

Legend



Queue: H

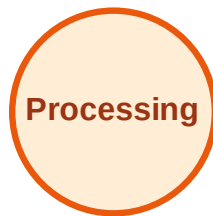
Visited: A, B, C, D, E, F, G, J



BFS Traversal

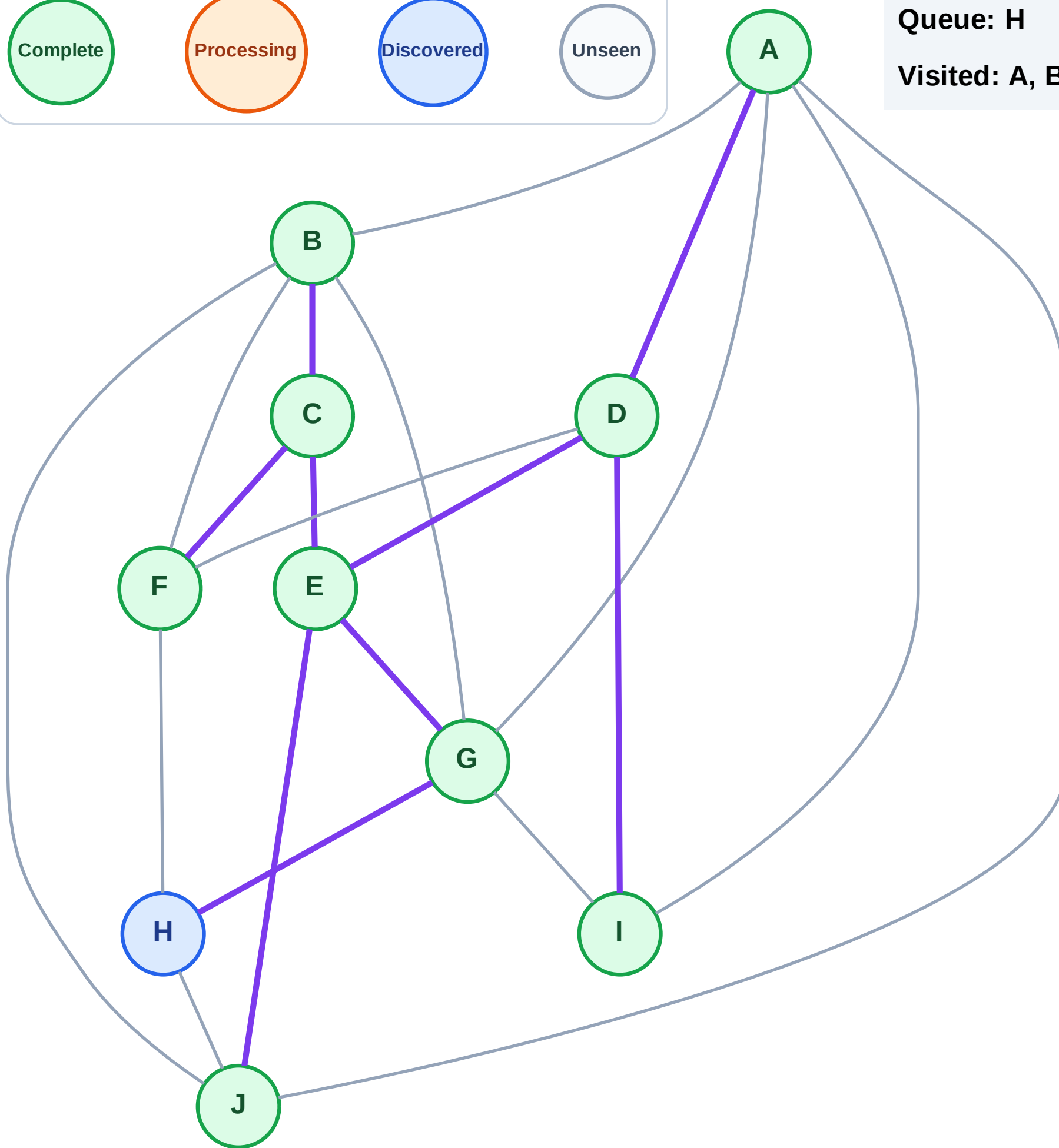
Step 19: No new neighbors from I

Legend



Queue: H

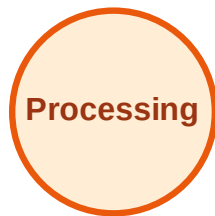
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

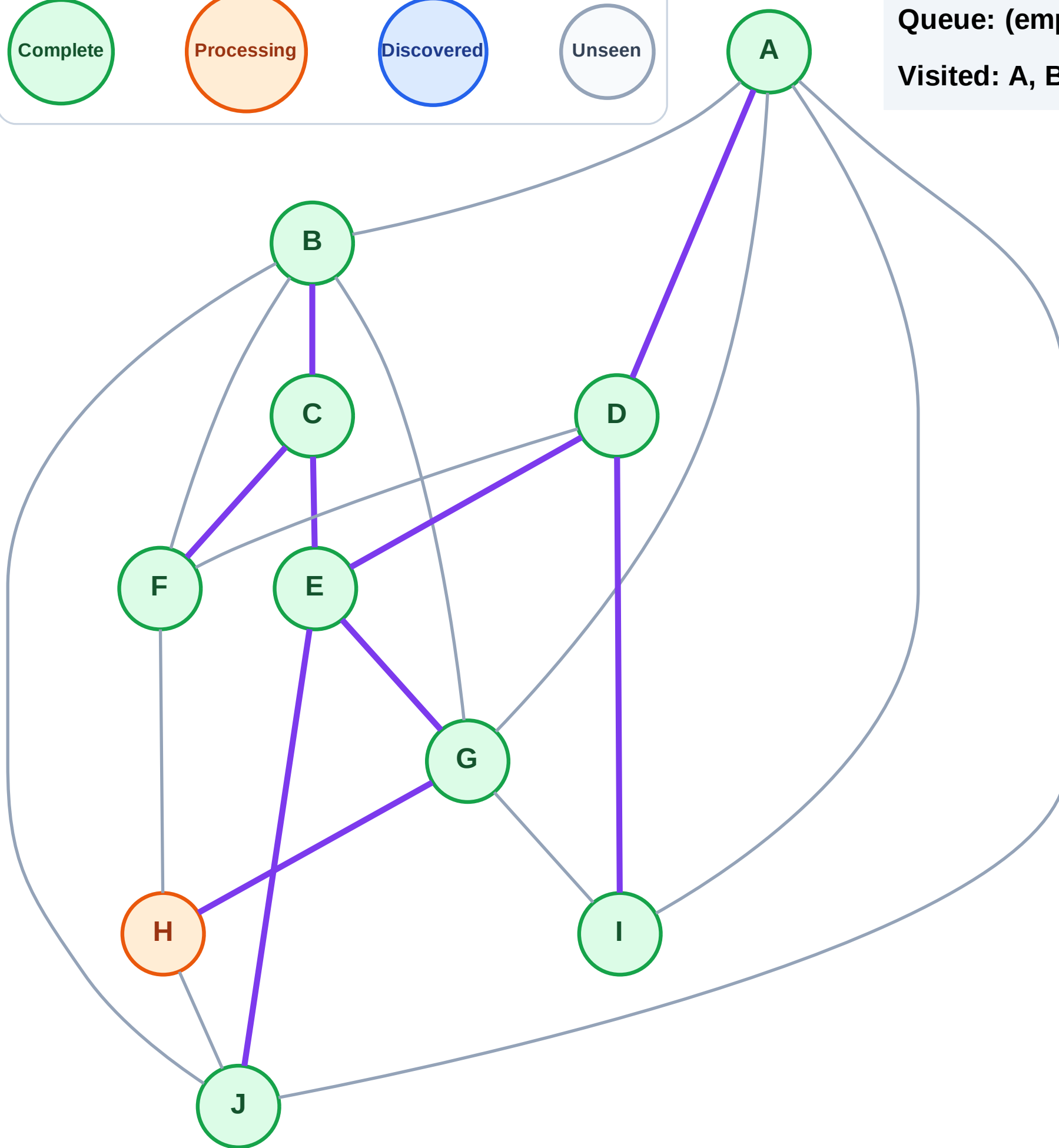
Step 20: Process node H

Legend



Queue: (empty)

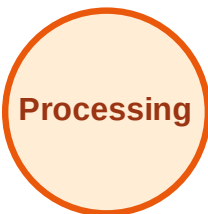
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

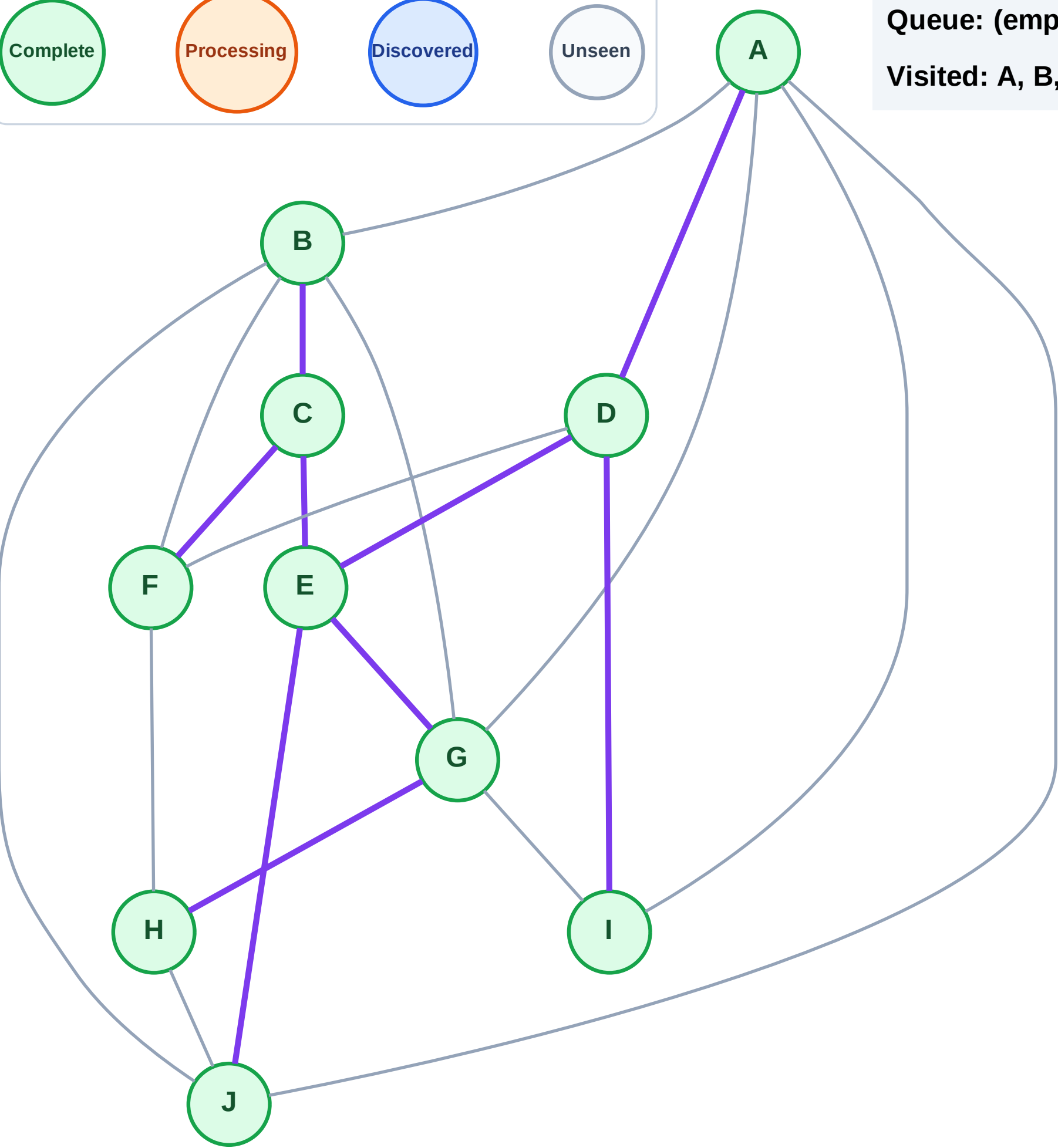
Step 21: No new neighbors from H

Legend



Queue: (empty)

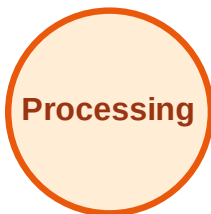
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

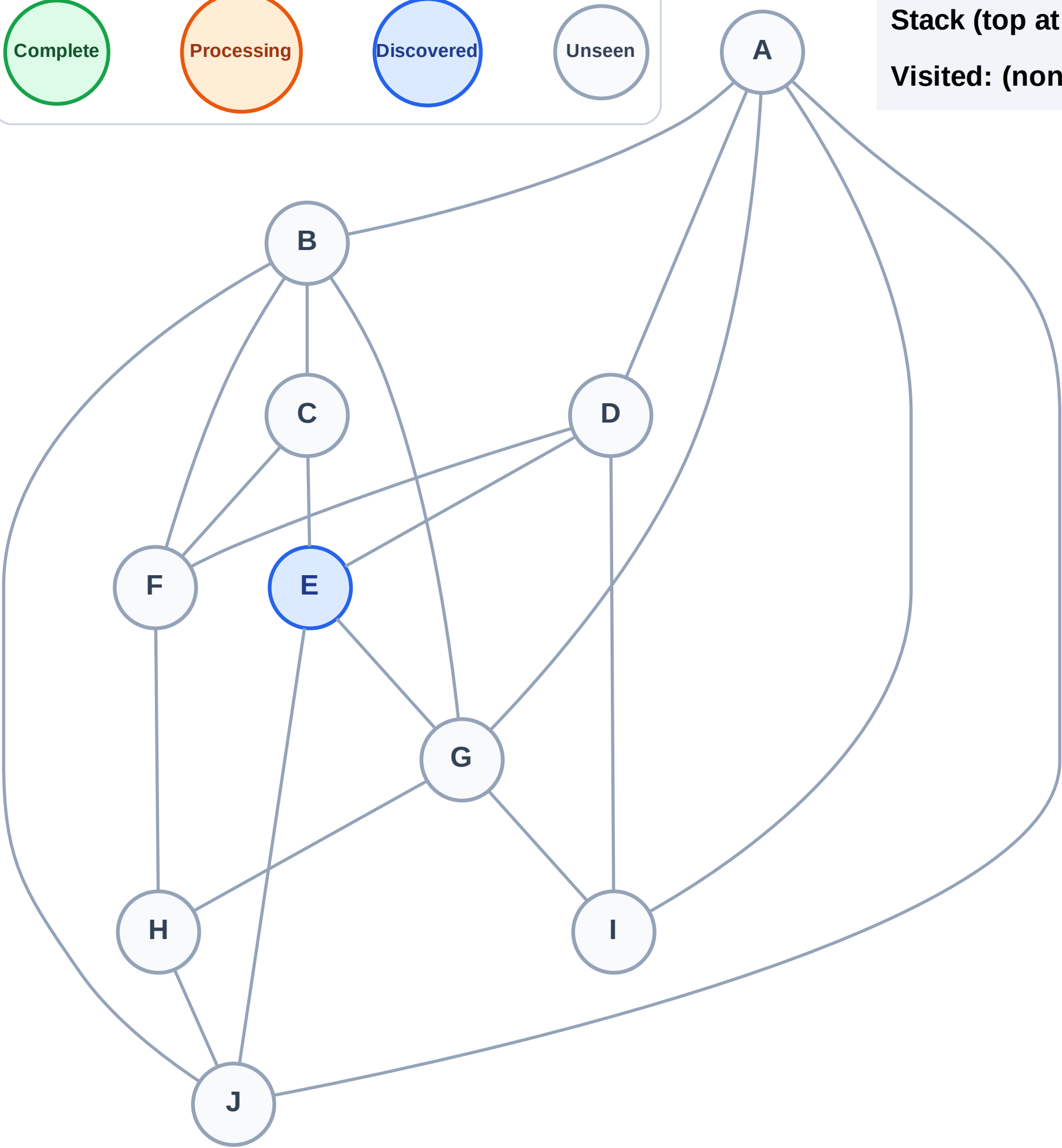
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

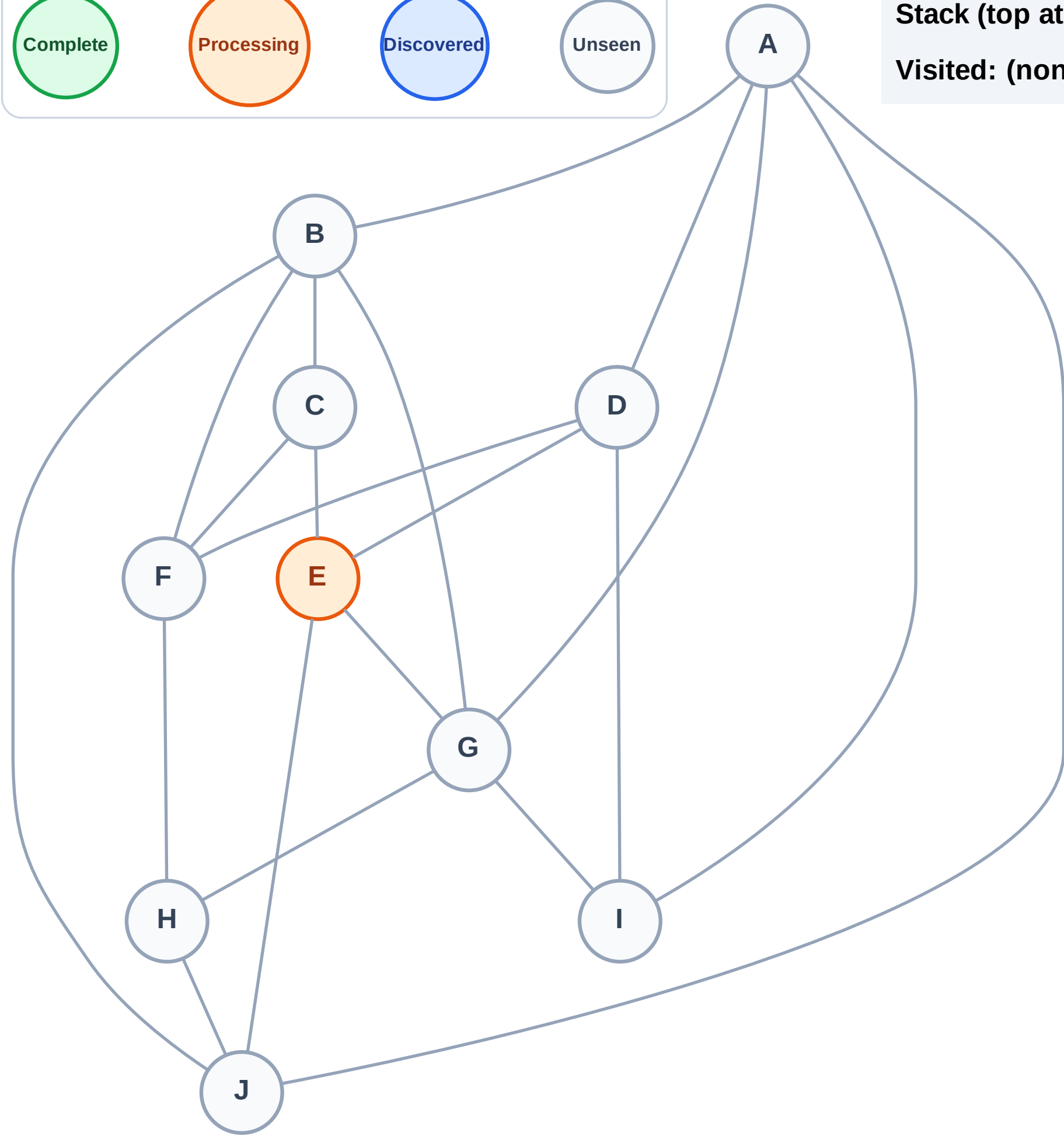
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover J, G, D, C from E

Legend

Complete

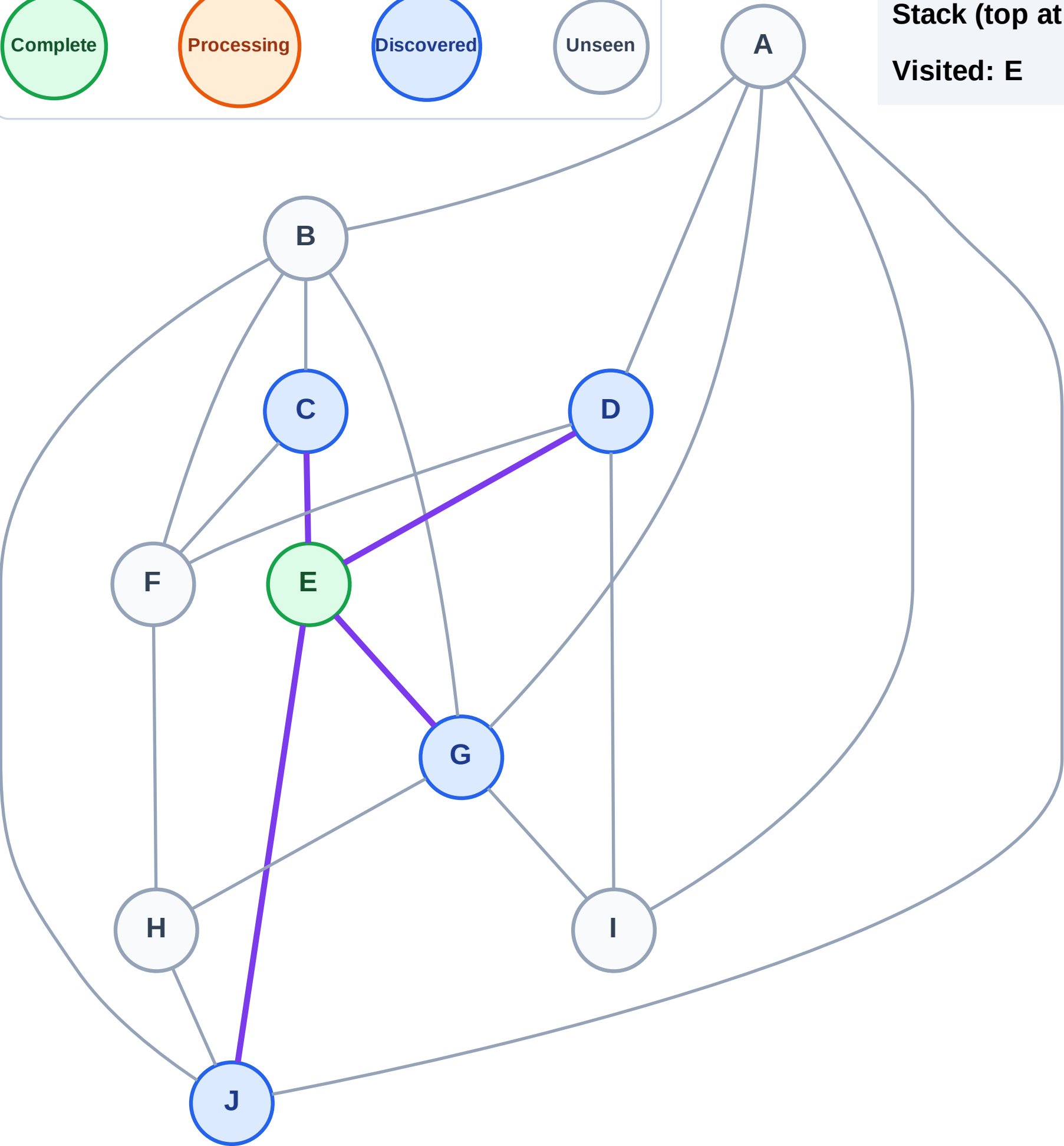
Processing

Discovered

Unseen

Stack (top at right): J | G | D | C

Visited: E



DFS Traversal

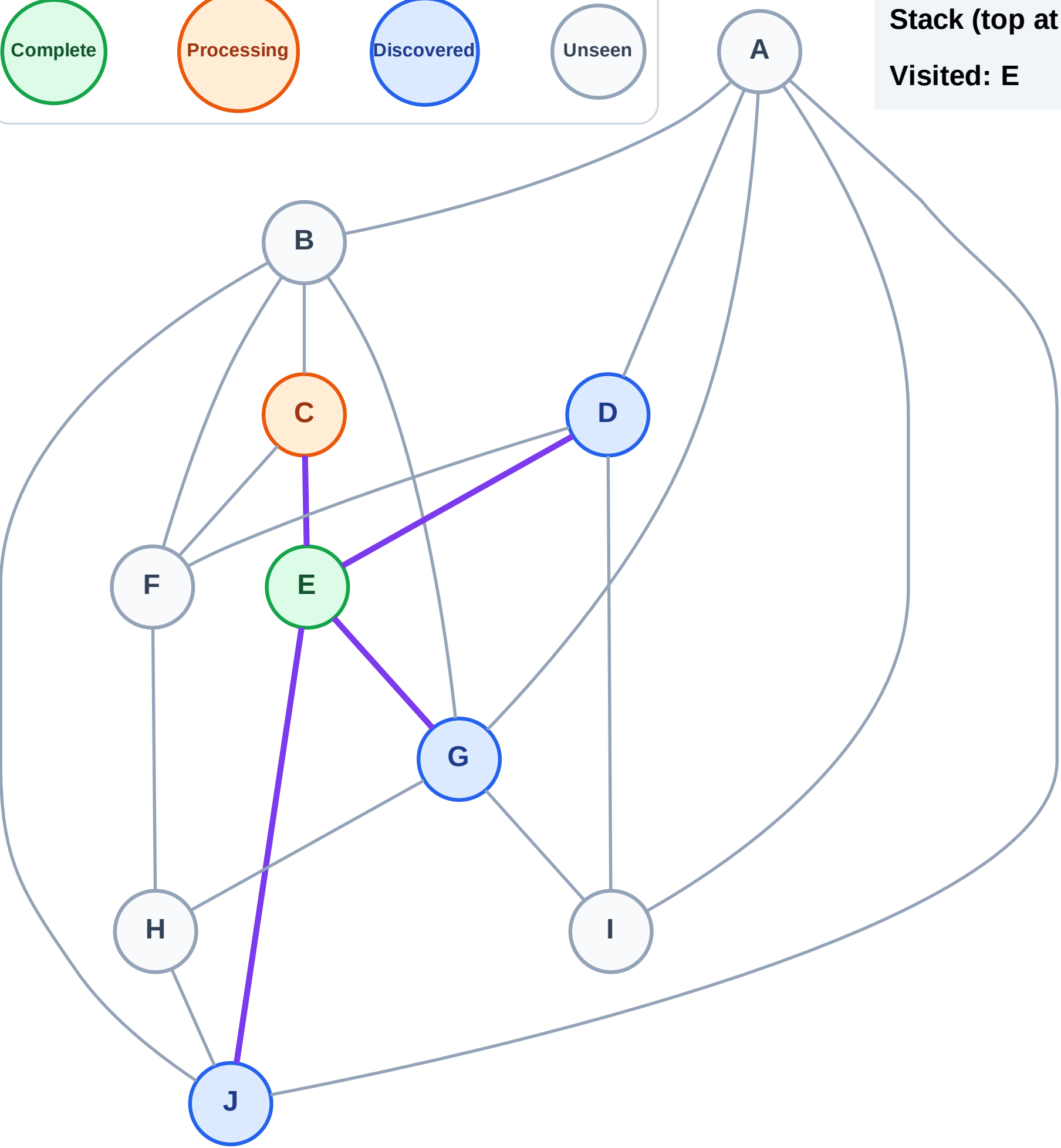
Step 4: Process node C

Legend



Stack (top at right): J | G | D

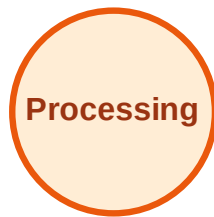
Visited: E



DFS Traversal

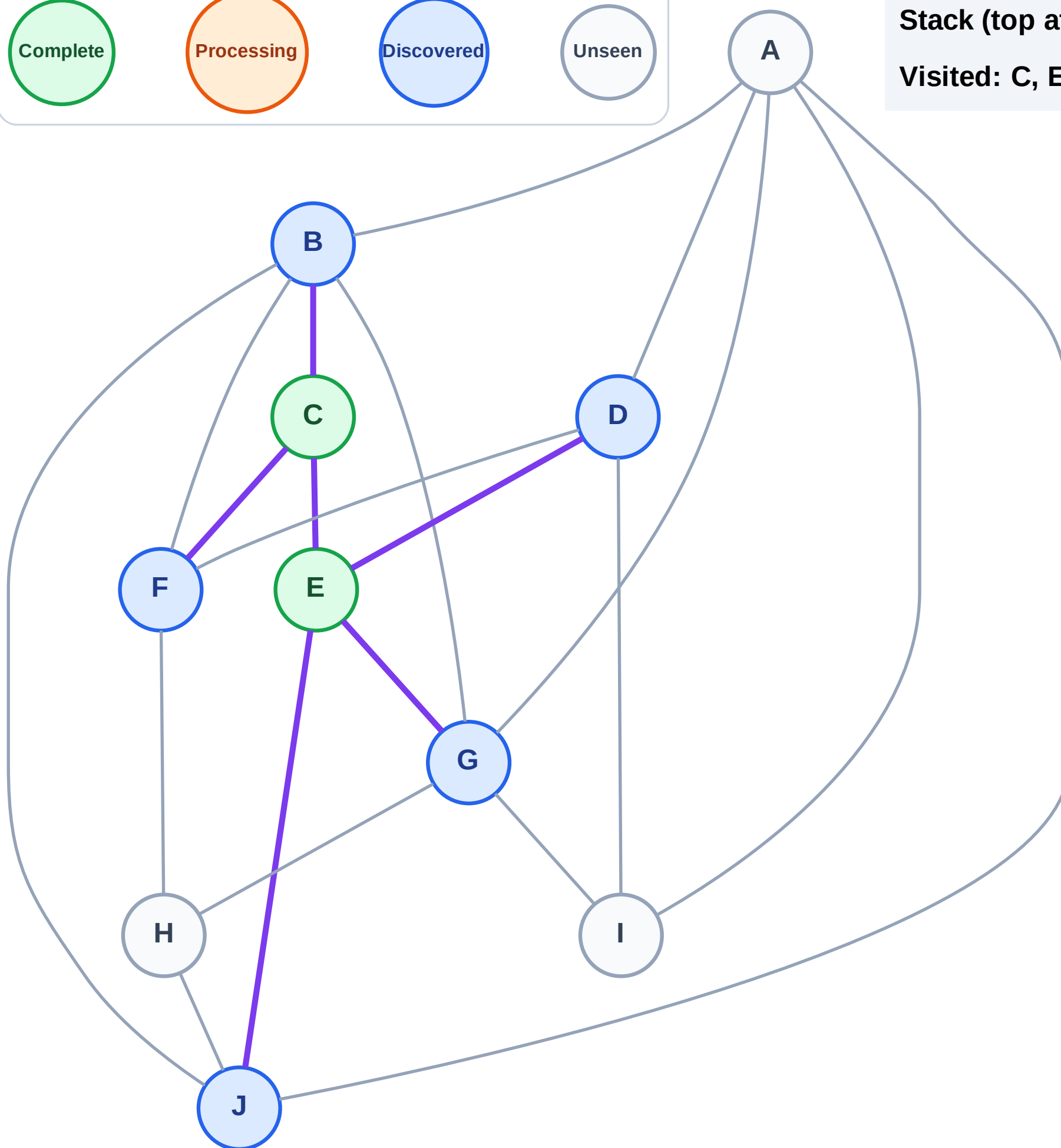
Step 5: Discover F, B from C

Legend



Stack (top at right): J | G | D | F | B

Visited: C, E



DFS Traversal

Step 6: Process node B

Legend

Complete

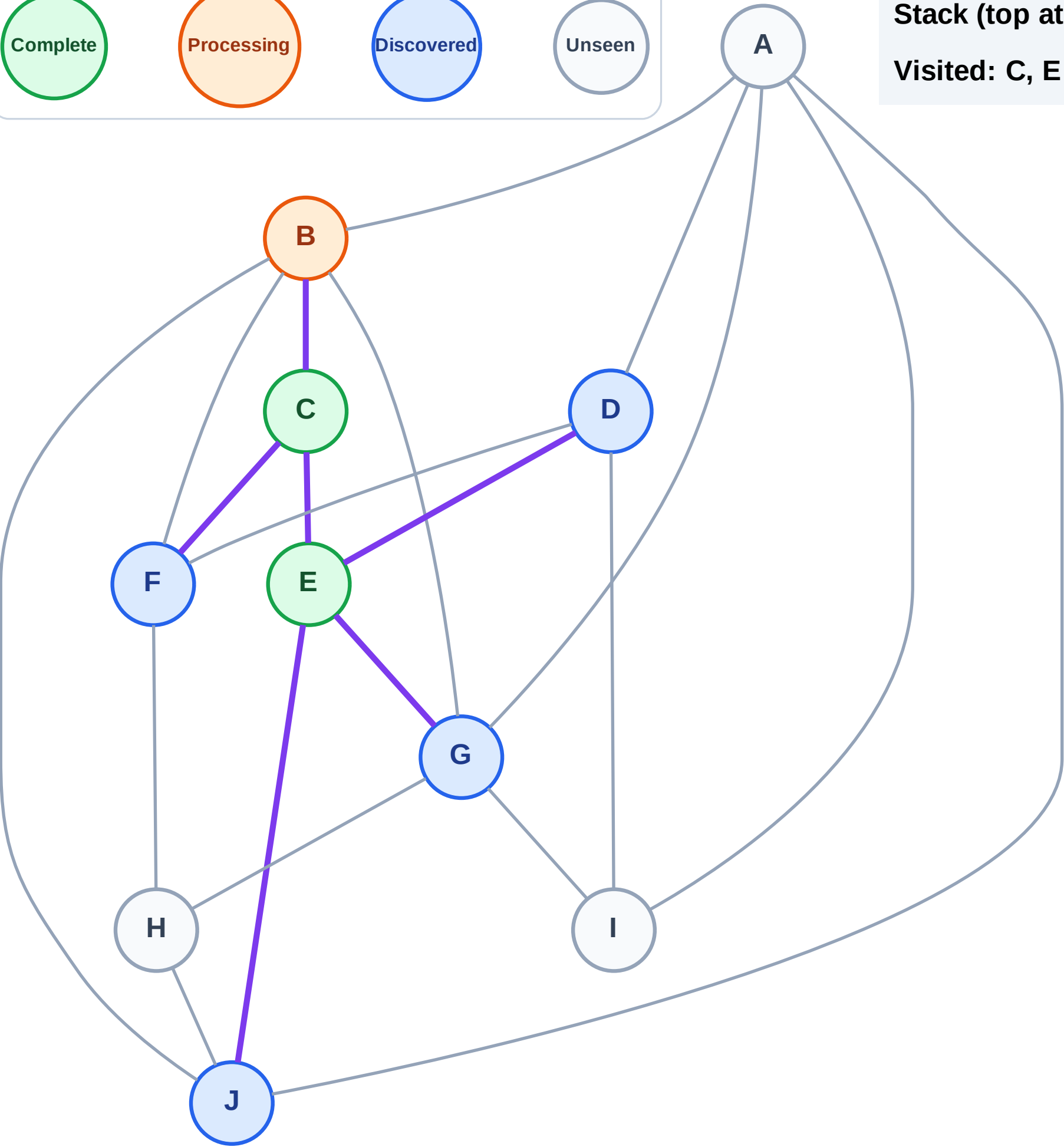
Processing

Discovered

Unseen

Stack (top at right): J | G | D | F

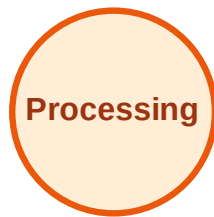
Visited: C, E



DFS Traversal

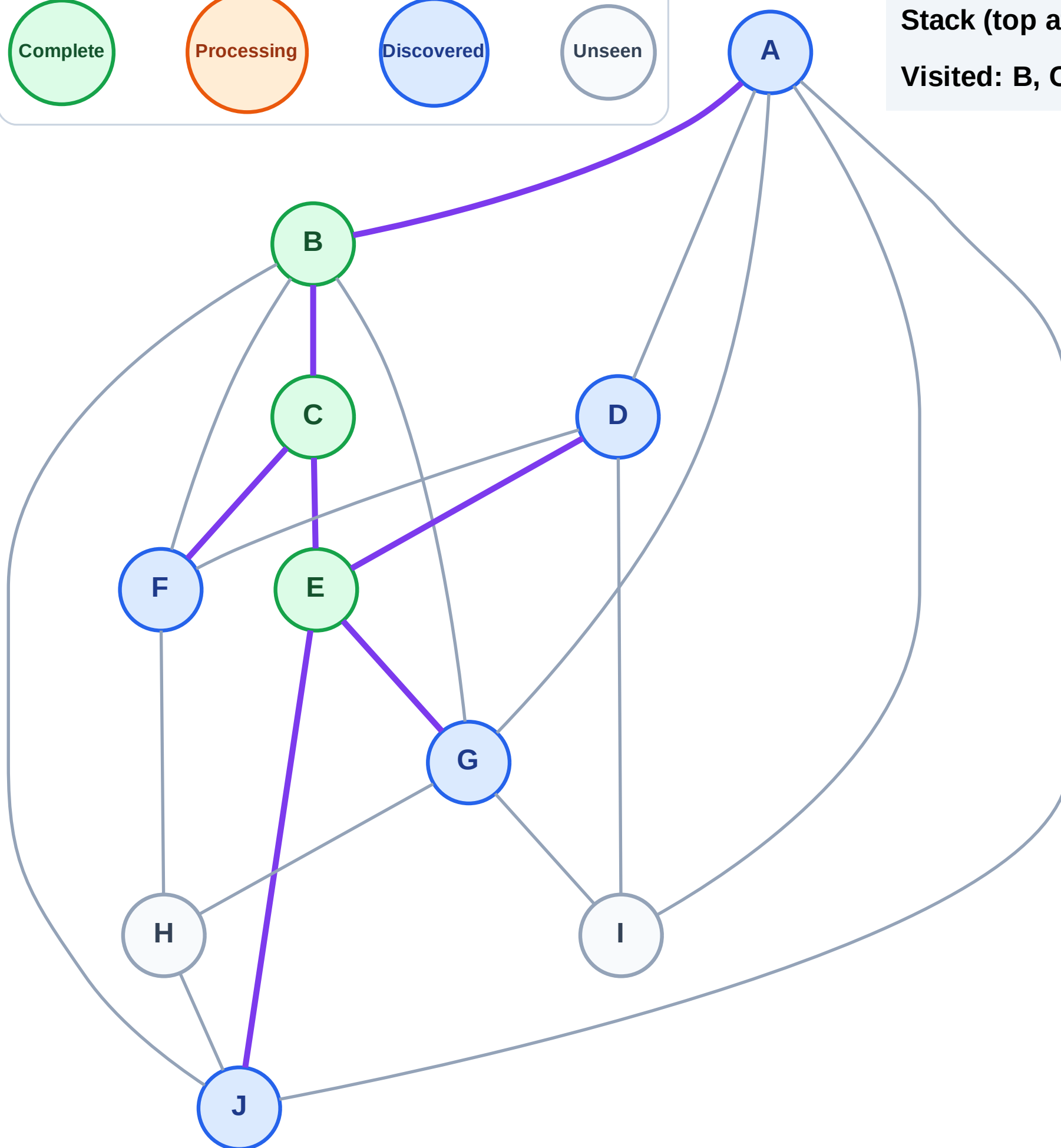
Step 7: Discover A from B

Legend



Stack (top at right): J | G | D | F | A

Visited: B, C, E



DFS Traversal

Step 8: Process node A

Legend

Complete

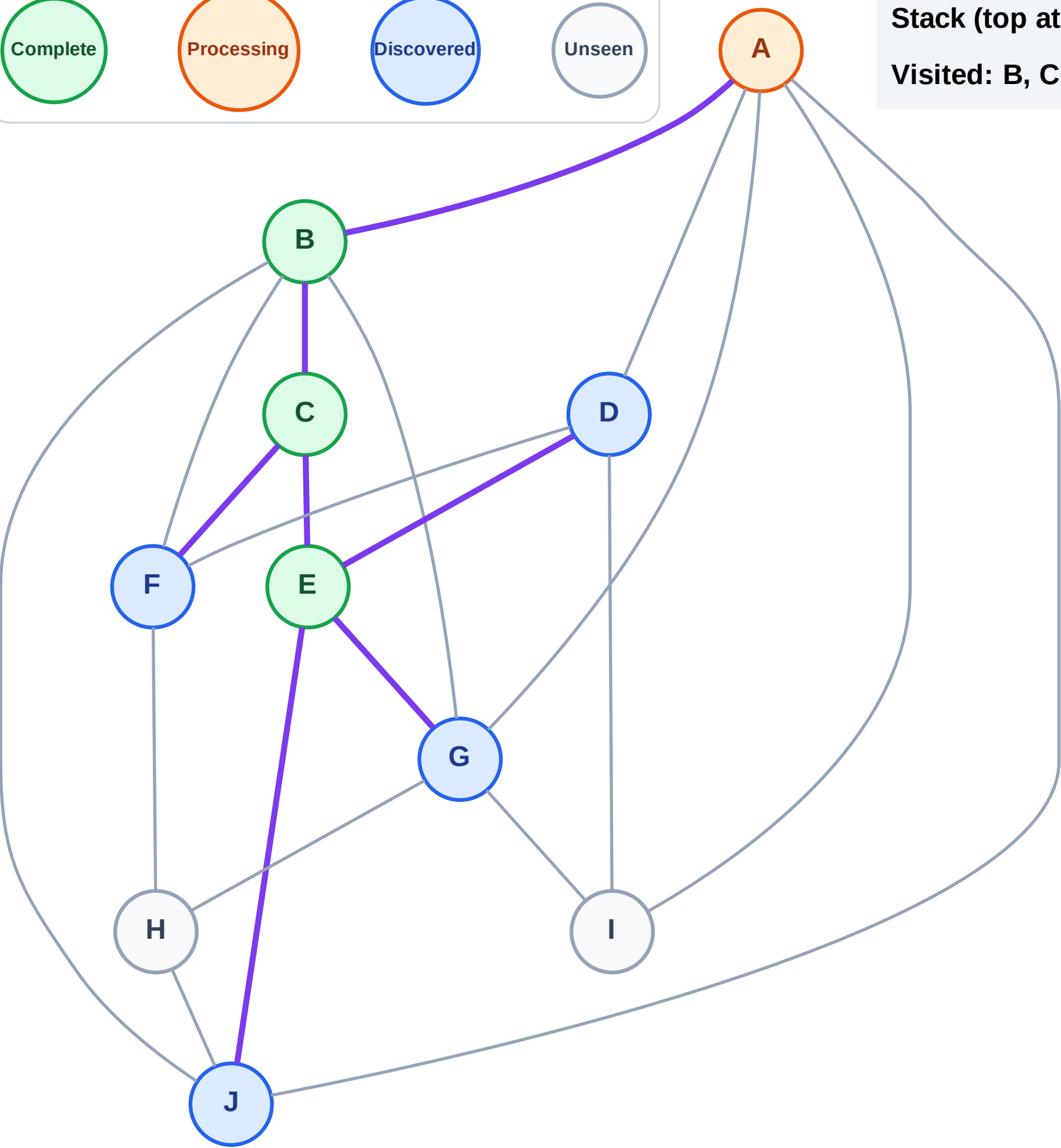
Processing

Discovered

Unseen

Stack (top at right): J | G | D | F

Visited: B, C, E



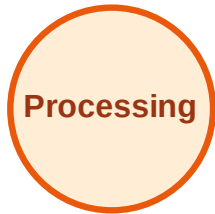
DFS Traversal

Step 9: Discover I from A

Legend



Complete



Processing



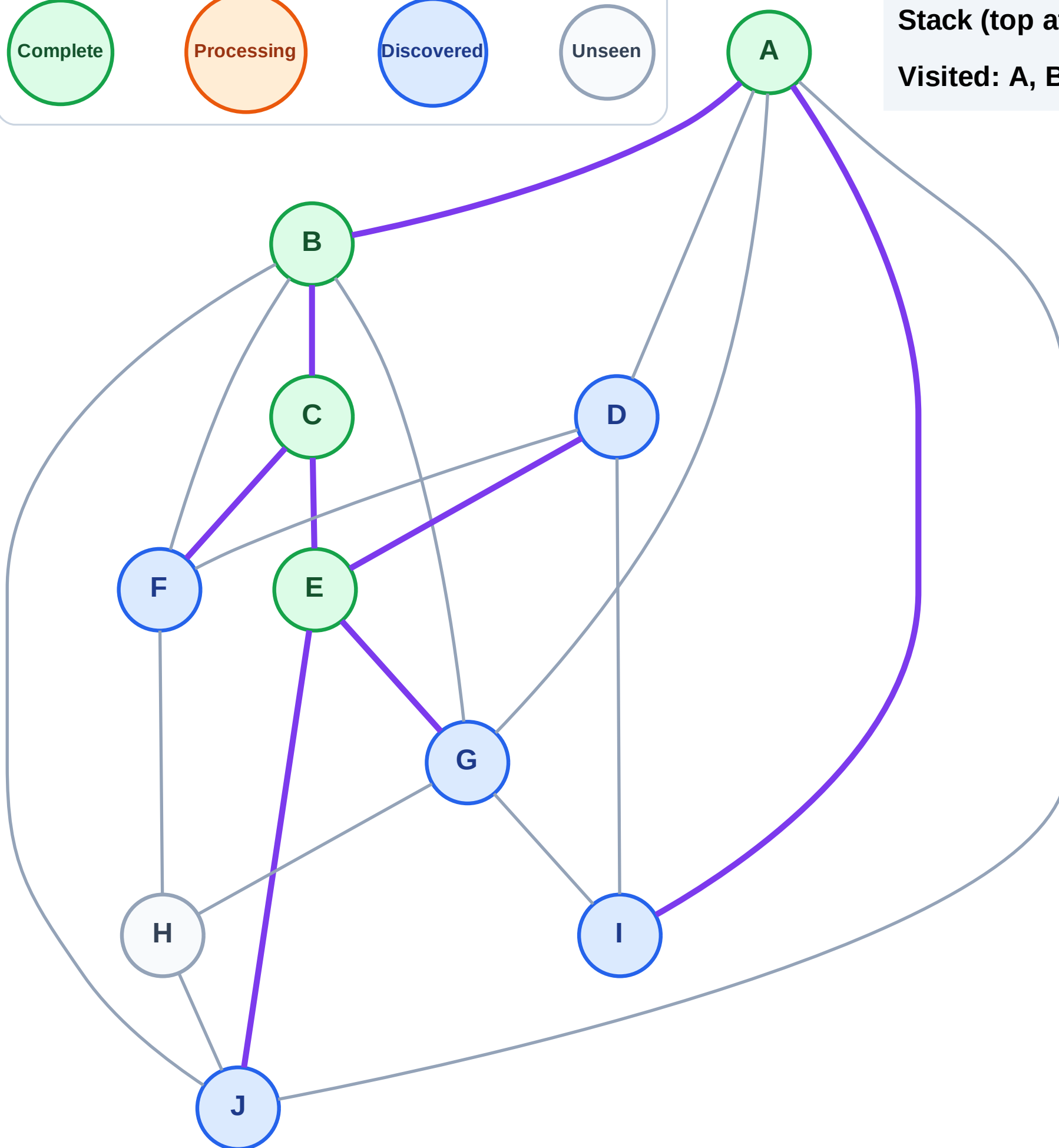
Discovered



Unseen

Stack (top at right): J | G | D | F | I

Visited: A, B, C, E



DFS Traversal

Step 10: Process node I

Legend



Complete



Processing



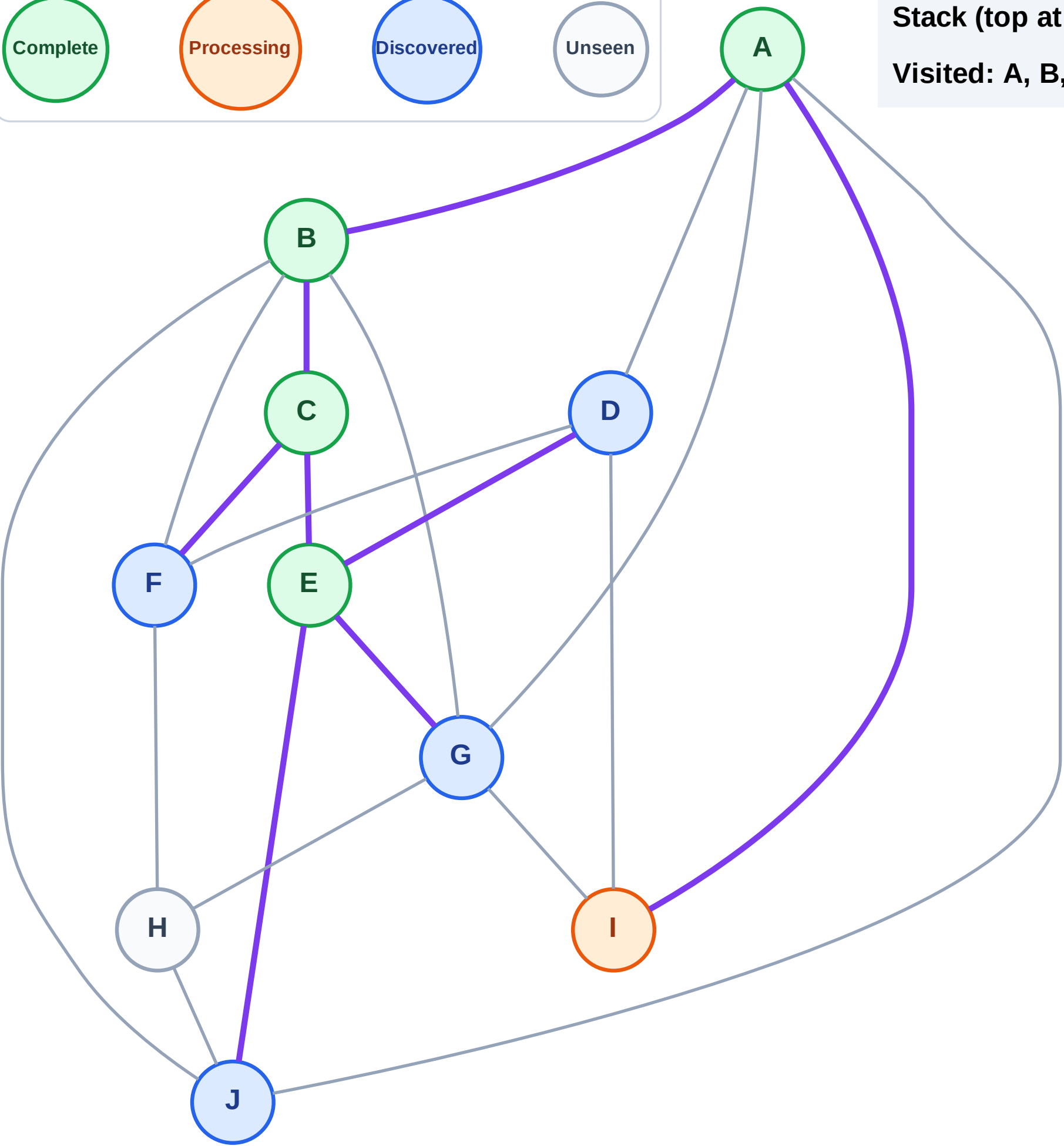
Discovered



Unseen

Stack (top at right): J | G | D | F

Visited: A, B, C, E



DFS Traversal

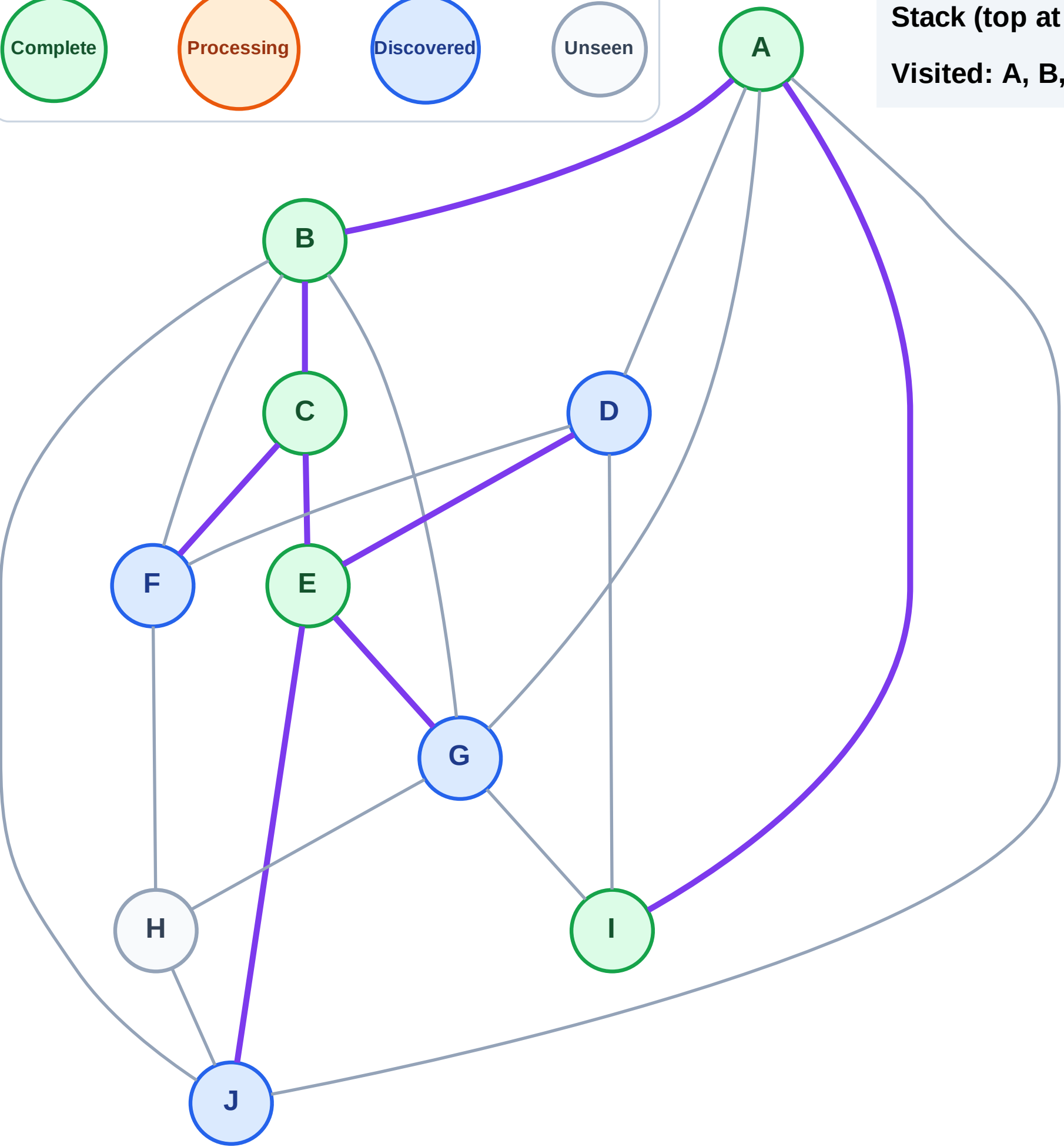
Step 11: No new neighbors from I

Legend



Stack (top at right): J | G | D | F

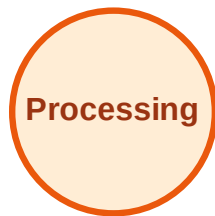
Visited: A, B, C, E, I



DFS Traversal

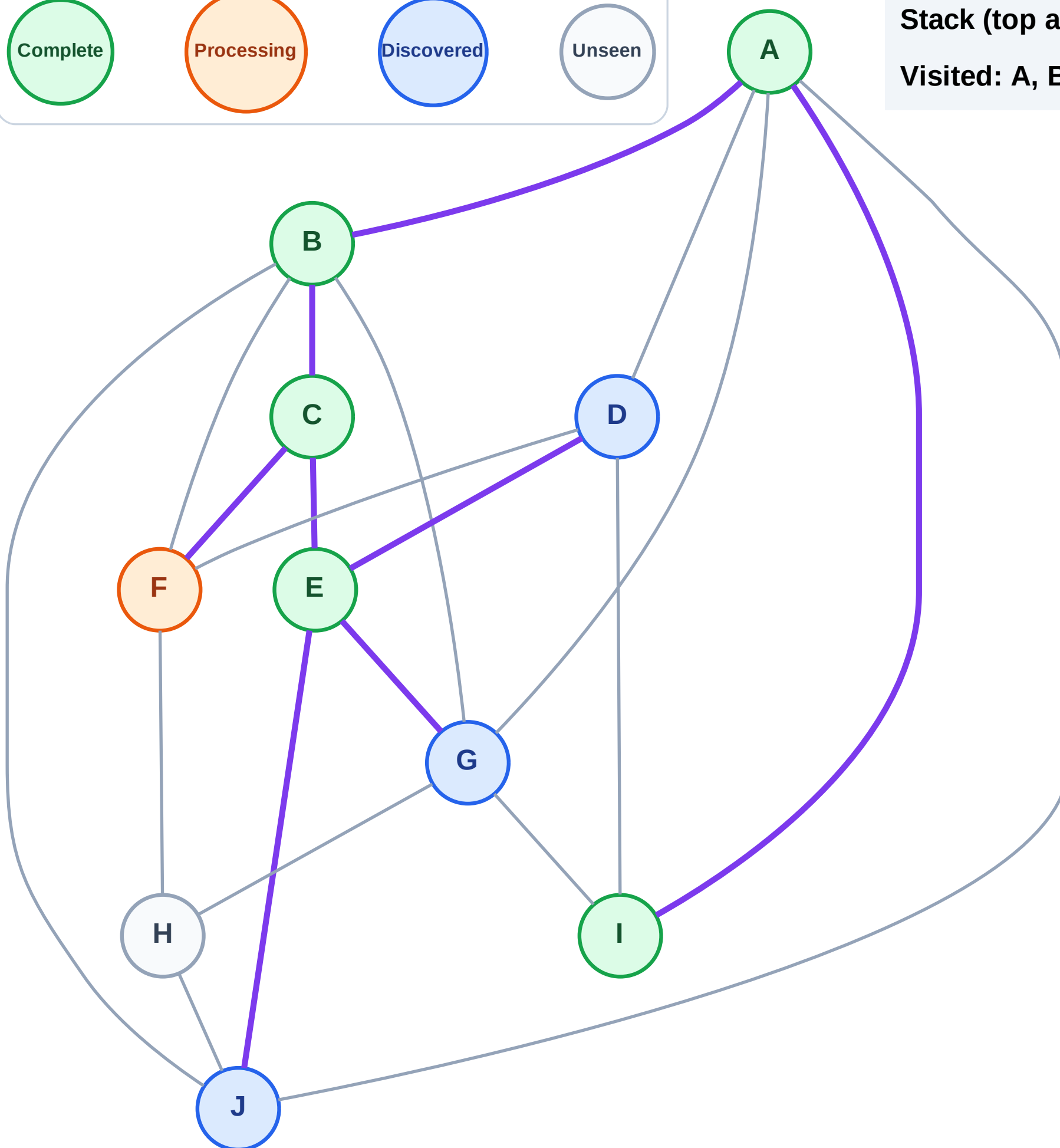
Step 12: Process node F

Legend



Stack (top at right): J | G | D

Visited: A, B, C, E, I



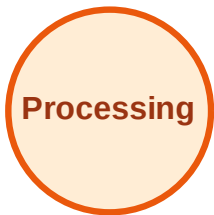
DFS Traversal

Step 13: Discover H from F

Legend



Complete



Processing



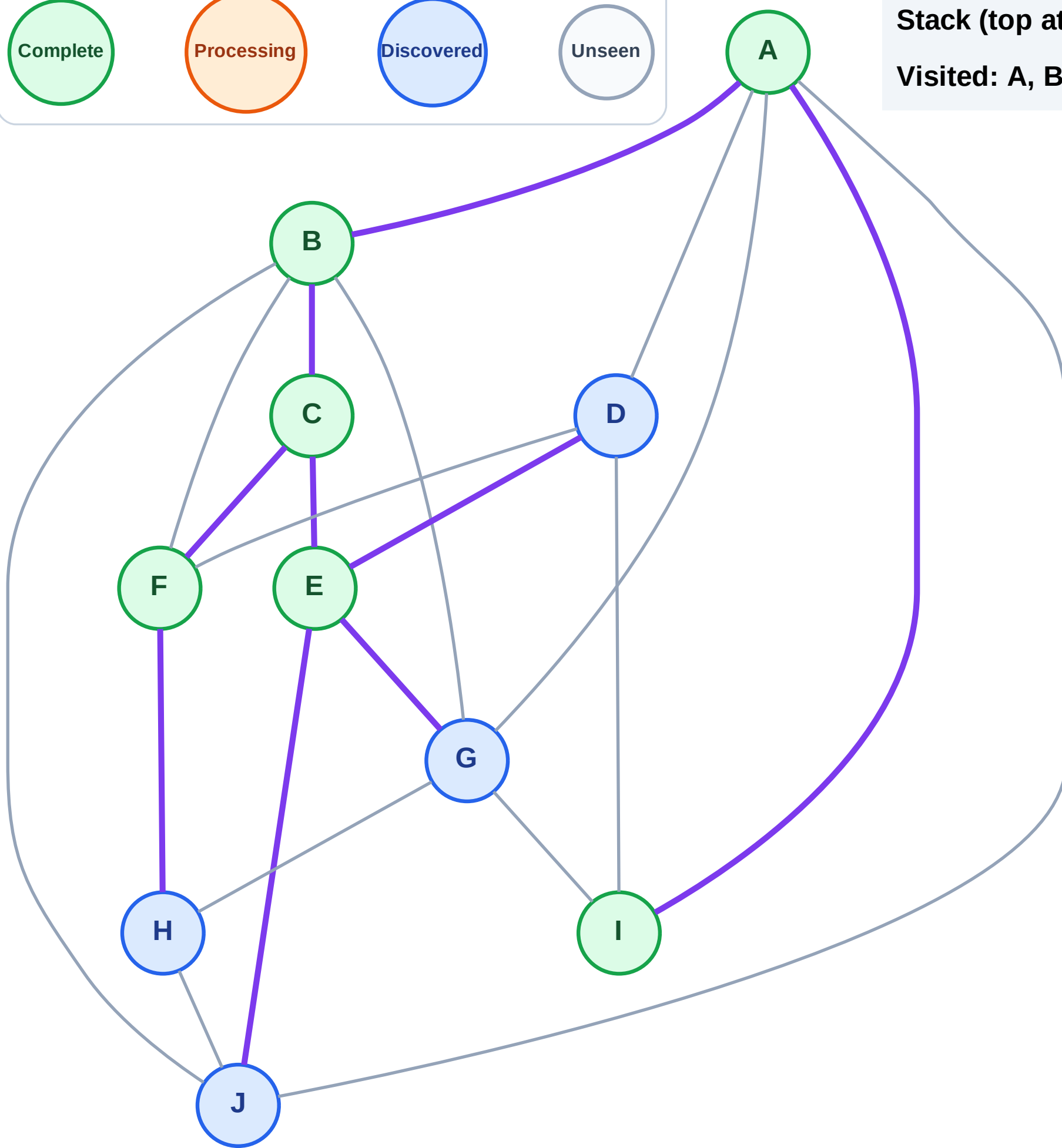
Discovered



Unseen

Stack (top at right): J | G | D | H

Visited: A, B, C, E, F, I



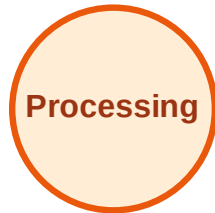
DFS Traversal

Step 14: Process node H

Legend



Complete



Processing



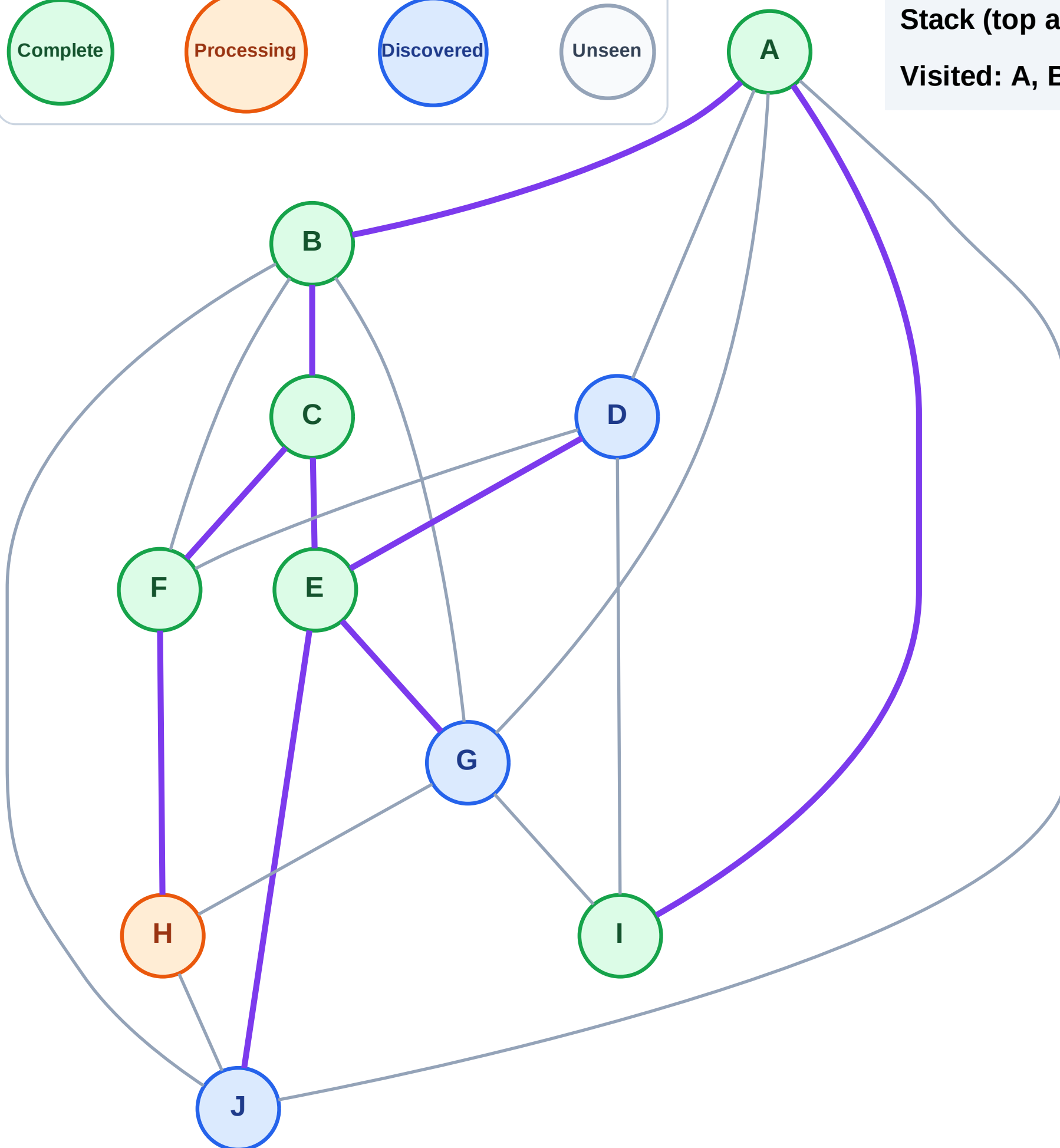
Discovered



Unseen

Stack (top at right): J | G | D

Visited: A, B, C, E, F, I



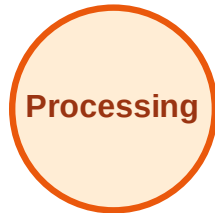
DFS Traversal

Step 15: No new neighbors from H

Legend



Complete



Processing



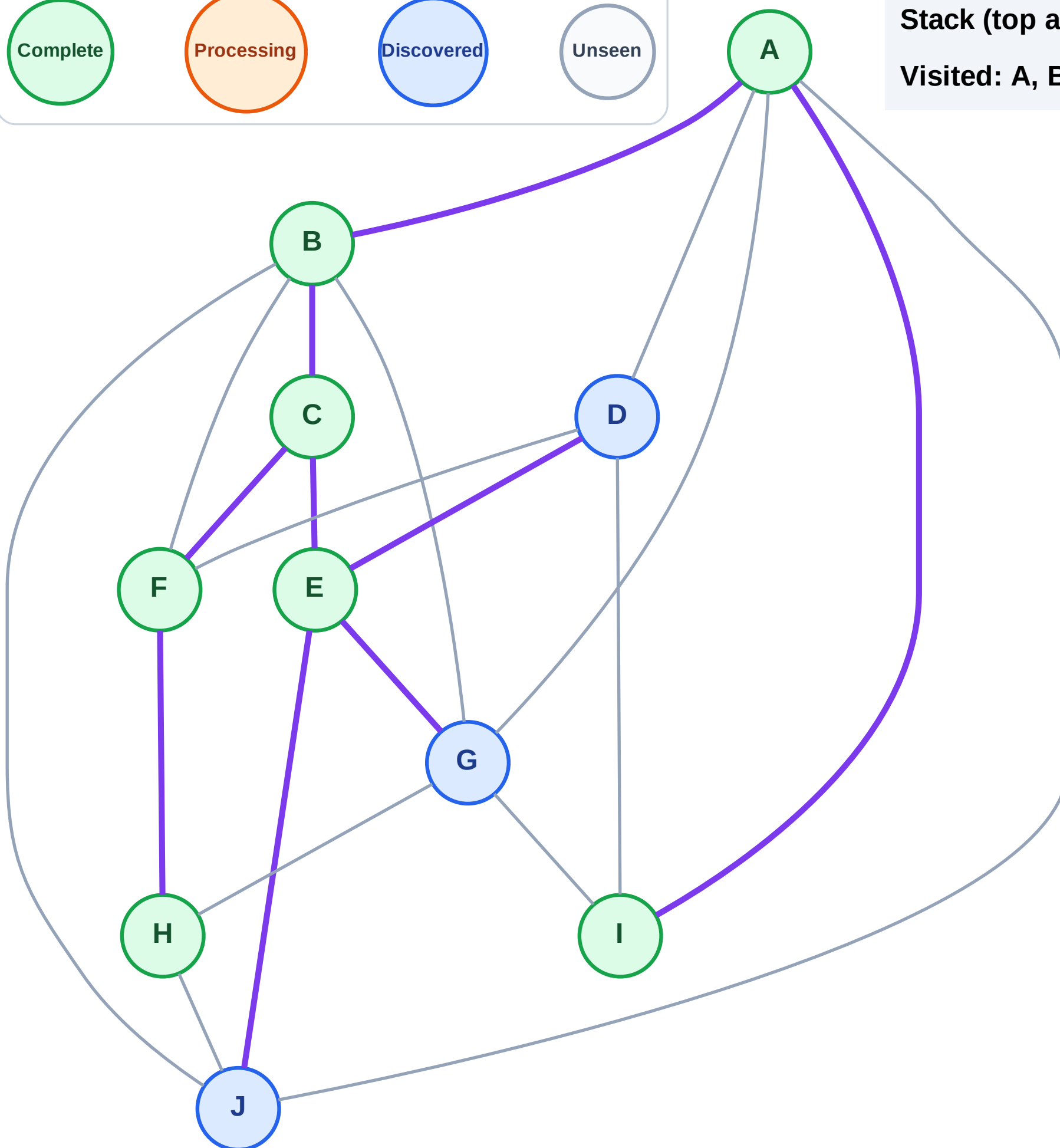
Discovered



Unseen

Stack (top at right): J | G | D

Visited: A, B, C, E, F, H, I



DFS Traversal

Step 16: Process node D

Legend

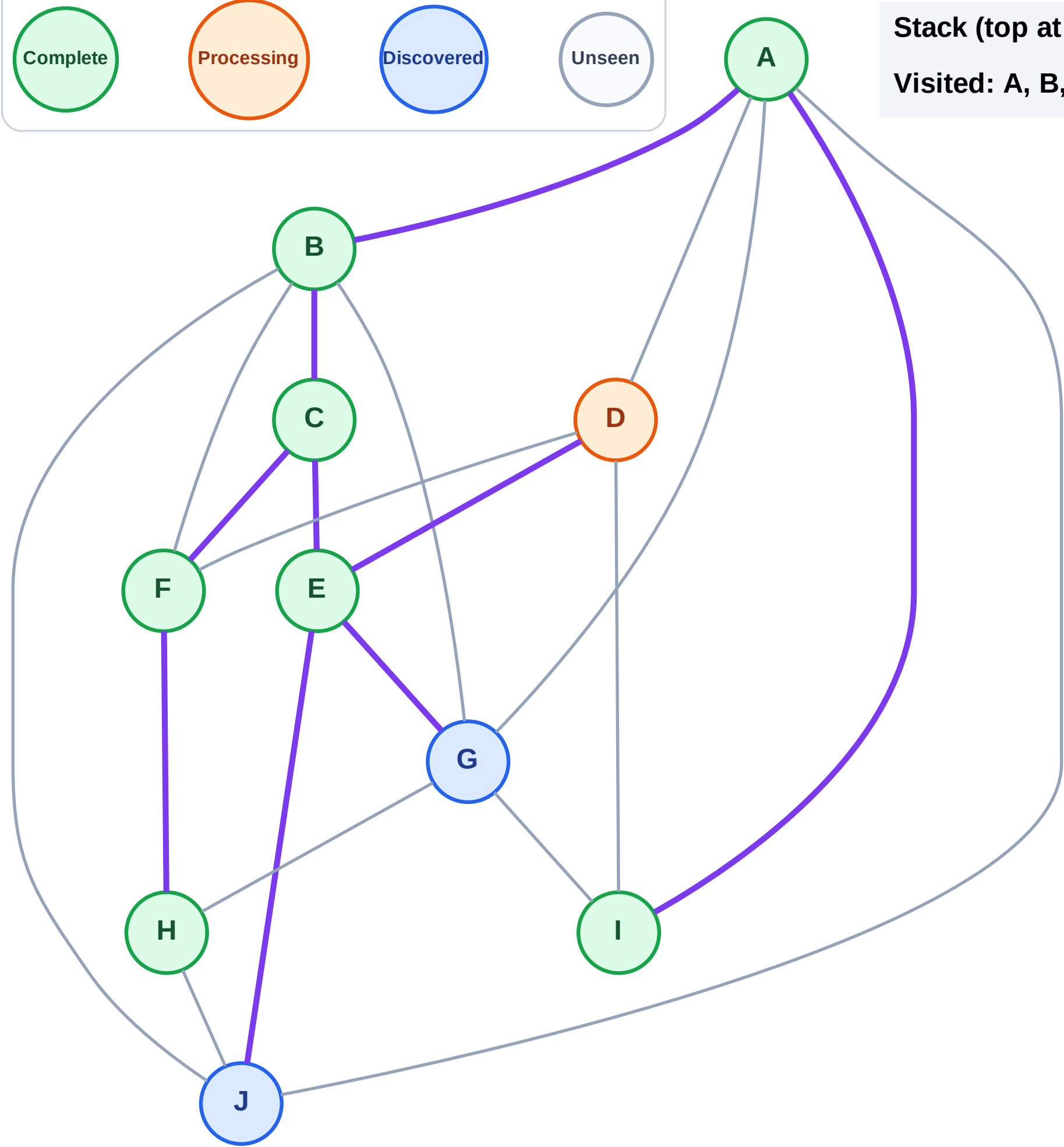
Complete

Processing

Discovered

Unseen

Stack (top at right): J | G
Visited: A, B, C, E, F, H, I



DFS Traversal

Step 17: No new neighbors from D

Legend

Complete

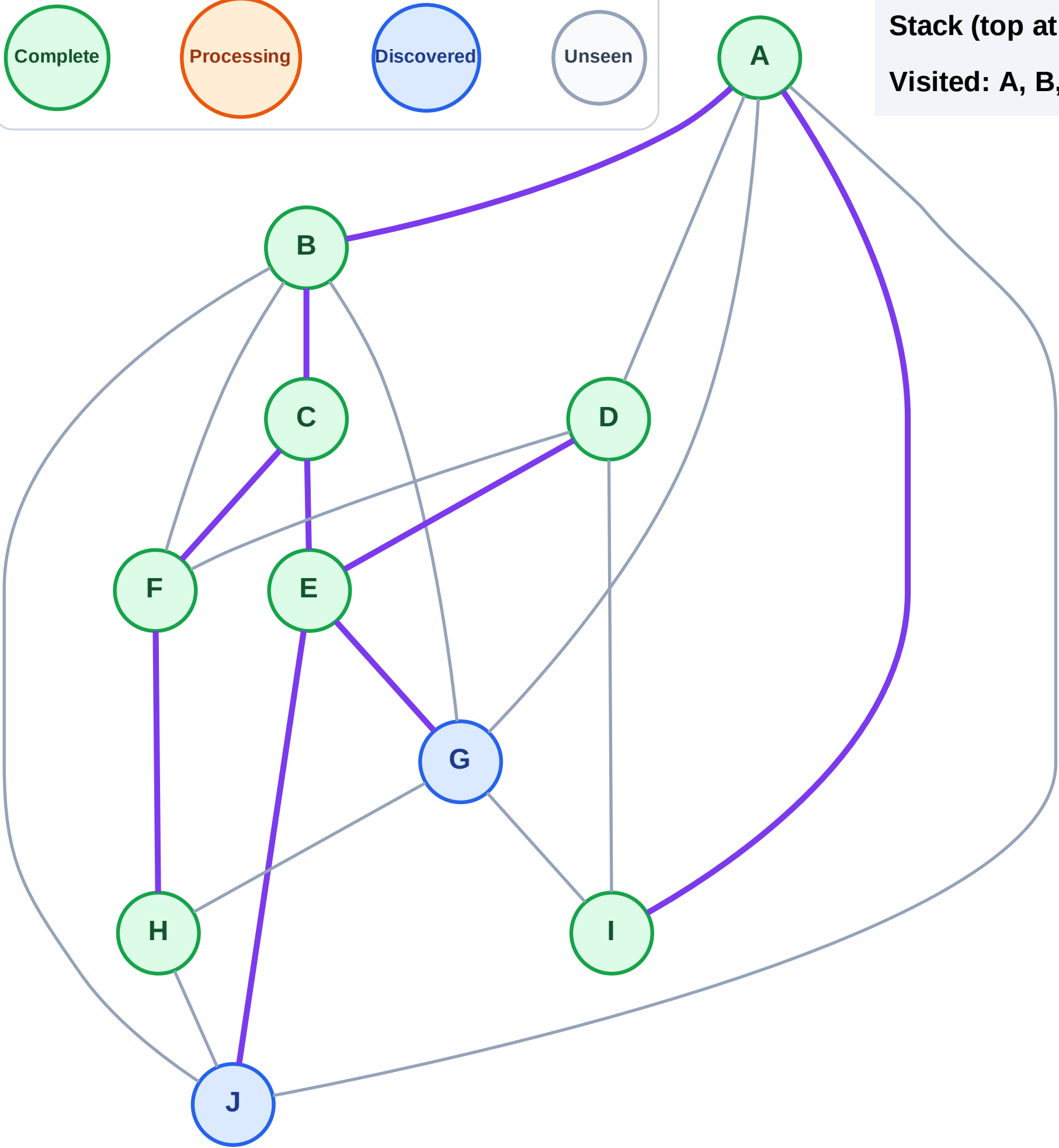
Processing

Discovered

Unseen

Stack (top at right): J | G

Visited: A, B, C, D, E, F, H, I



DFS Traversal

Step 18: Process node G

Legend

Complete

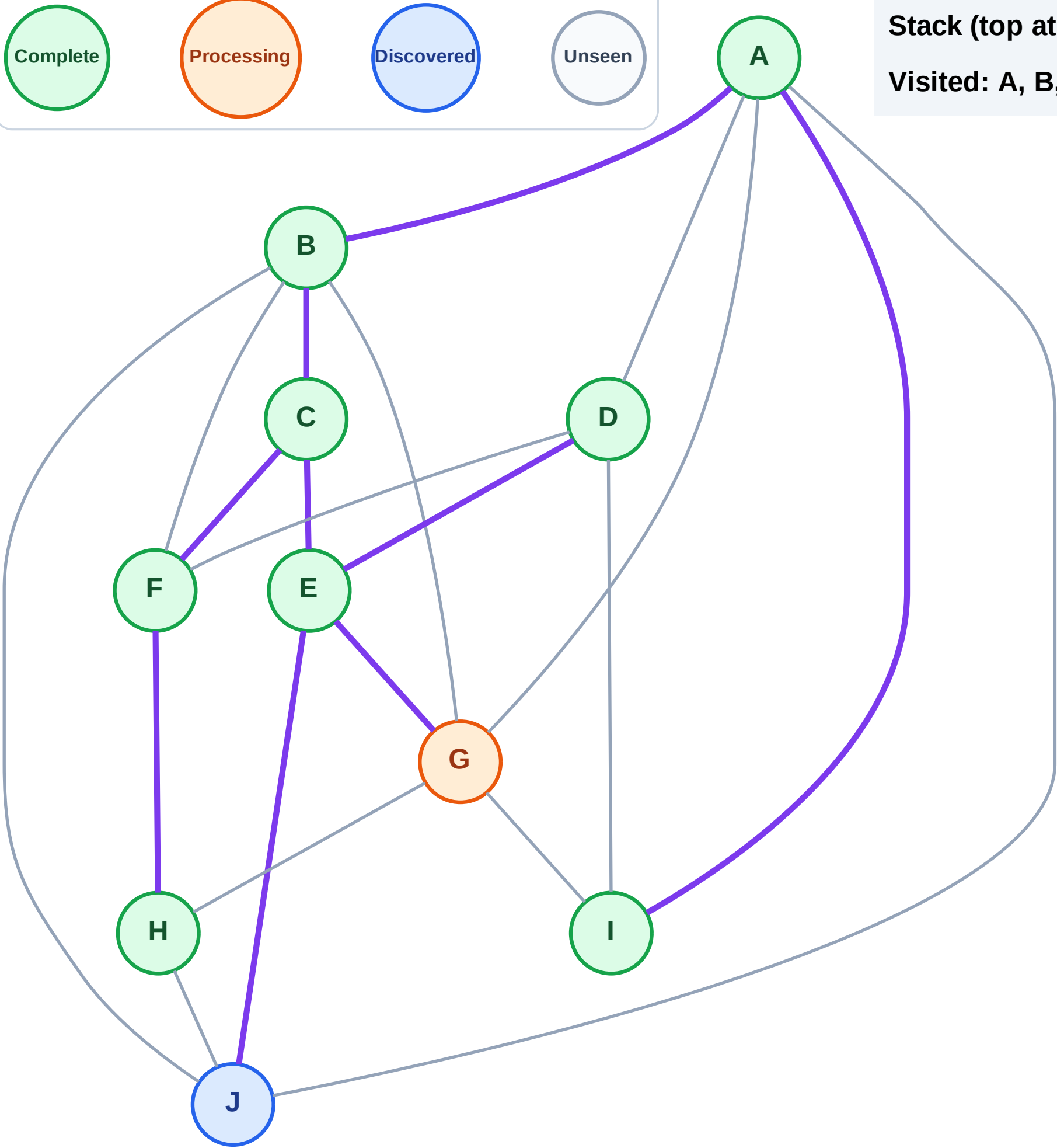
Processing

Discovered

Unseen

Stack (top at right): J

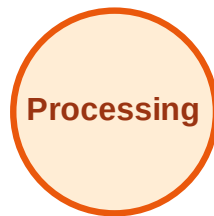
Visited: A, B, C, D, E, F, H, I



DFS Traversal

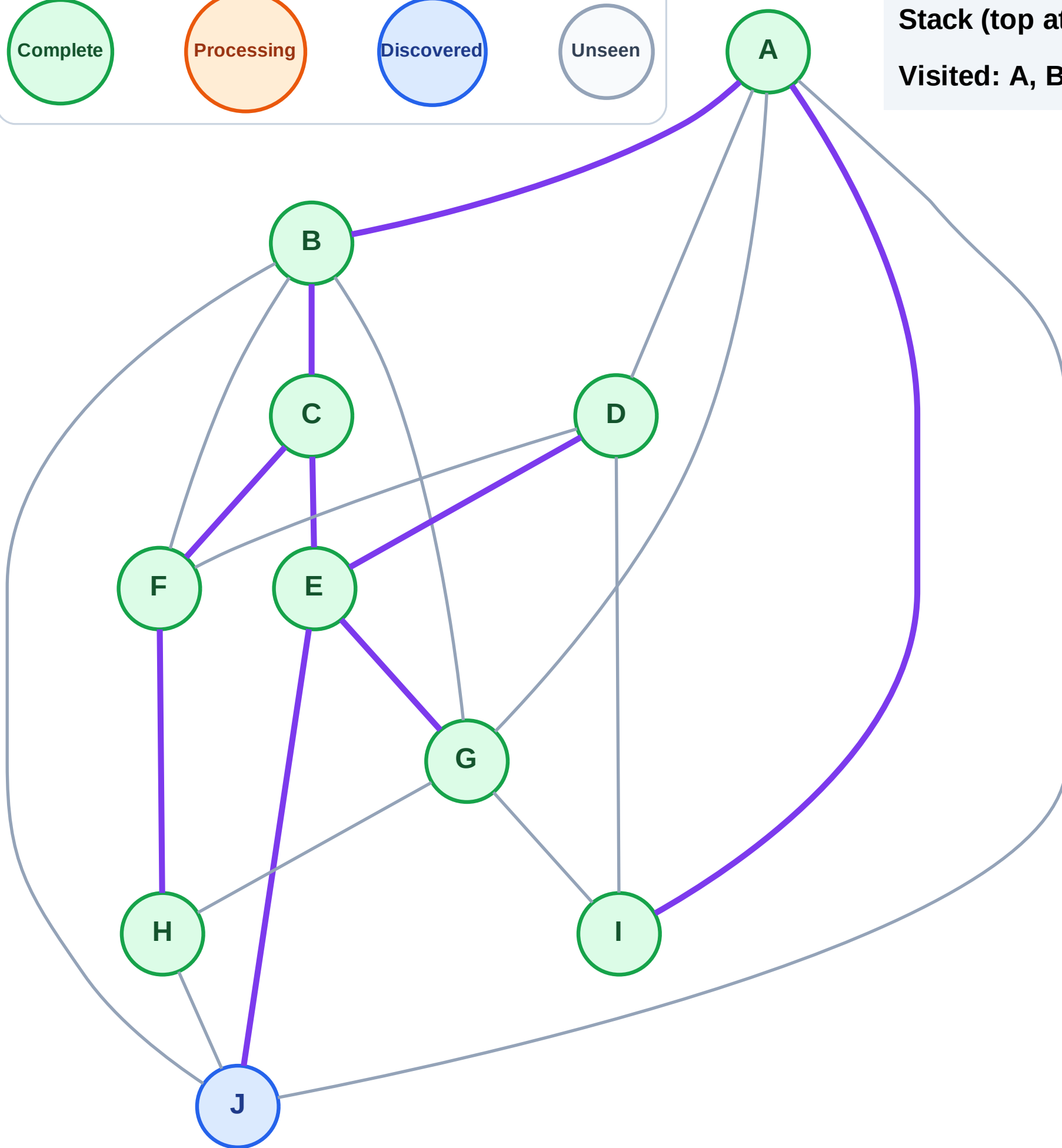
Step 19: No new neighbors from G

Legend



Stack (top at right): J

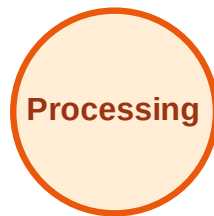
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

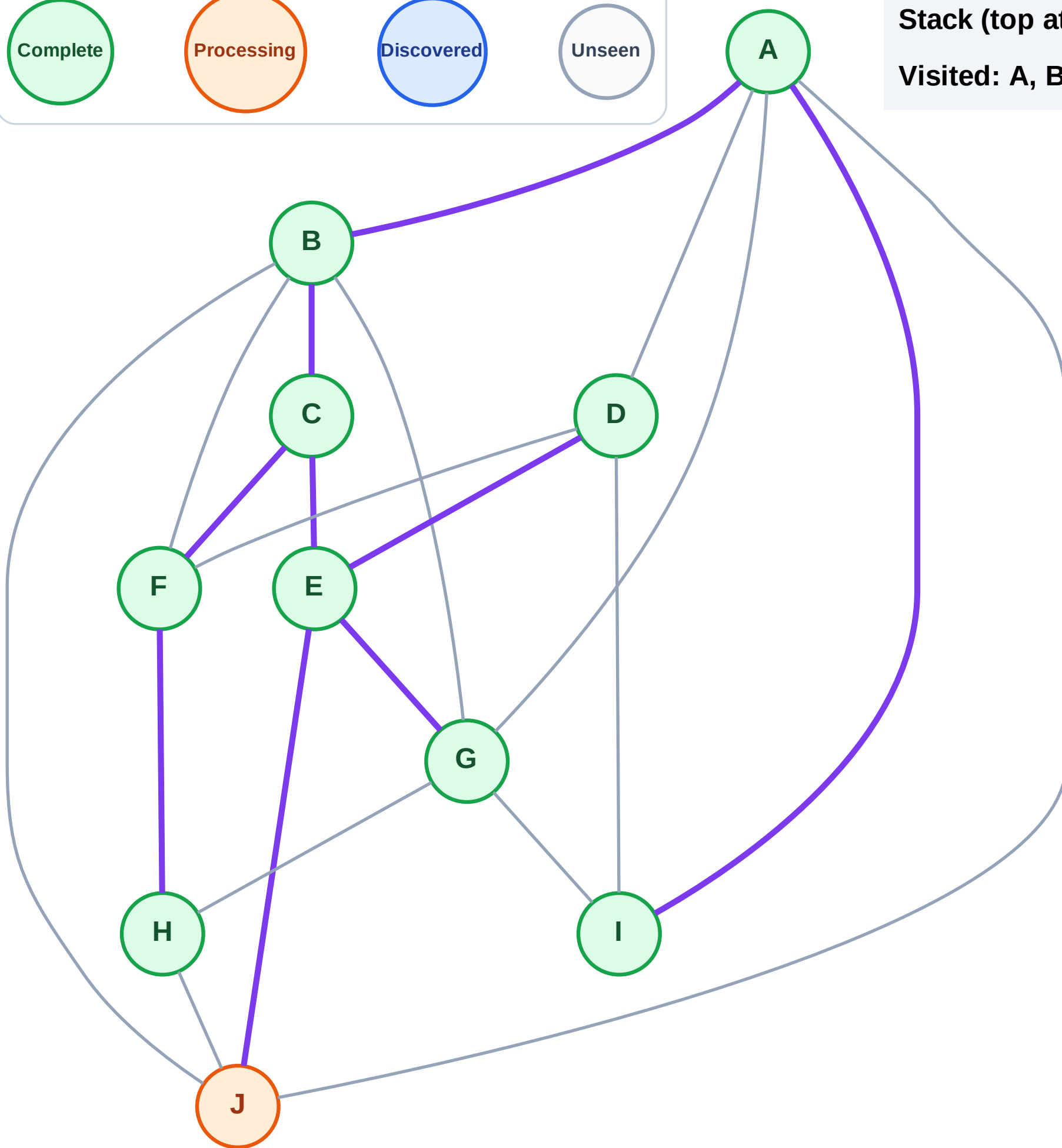
Step 20: Process node J

Legend



Stack (top at right): (empty)

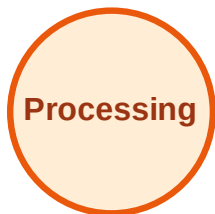
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

